

Philosophical Propaedeutic

in

Outline

by

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Adamant homines scientias et contemplationes particulares, aut quia autores et inventores se earum credunt, aut quia plurimum in illis operæ posuerunt, iisque maxime assueverunt. Hujusmodi vero homines, si ad philosophiam et contemplationes universales se contulerint, illas ex prioribus phantasiis detorquent et corrumpunt.

FRANC. BACO.

[Men grow attached to particular sciences and speculations, either because they fancy themselves their authors and inventors, or because they have bestowed the greatest labour upon them and become most habituated to them. But if men of this sort betake themselves to philosophy and universal speculations, they distort and corrupt these out of their former fancies. — Francis Bacon.]

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Introduction.

The philosophical propaedeutic sets itself to answer two fundamental questions: *what is philosophy? what validity can be ascribed to philosophy as a science?*

To answer these questions the propaedeutic must characterize philosophy by pointing out the shapes which it has assumed at its various stages of development; the general fundamental concept will then be determined more closely as it runs through significant stages. The propaedeutic, in letting philosophy come to be for contemplation in its own way, must everywhere take up something historically given. But, although the historical is made use of, we can here give neither an epitome of the history of philosophy in its entirety, nor a separate detailed treatment of individual, arbitrarily chosen principal parts on their own account; the historical must be grasped and used in such a way that, within the connection determined by the nature of the matter, it illuminates the tasks that arise, and serves the totality.

To characterize philosophy as a science in its relation to the other sciences, it is doubtless necessary to enter more closely into an investigation of that which brings about the distinctive character of the various sciences, and thus to point out the principle of division on which their separation essentially rests; but since the determination of what is universally scientific is and remains the main thing, the treatment of the mutual relation of the diverse sciences cannot take up the particulars of that ramification to such a degree that there should here be any question of an encyclopædia of the sciences.

Insofar as philosophy itself is a science articulated into definite members, it will likewise, for the sake of the fundamental concept, be necessary to point out, within this science's own domain, the inner connection of the larger divisions with the articulating principle; but since every survey in which the individual parts are not at the same time treated with some detail necessarily becomes superficial, the propaedeutic must, instead of setting itself the task of an encyclopædic presentation of the disciplines of philosophy, rather lay out its principal parts in such a way that, by a more detailed treatment of each division on its own, it may find occasion to illuminate the systematic structure from one or another definite side, and thereby gradually open an insight into the architectonic relations of the whole.

In order to comprise all the separate considerations under one comprehensive point of view, the propaedeutic begins with the conceptual determination that philosophy is a scientific theory of consciousness.

The theory of consciousness shows what it means to become conscious of something, and must to that extent distinguish between consciousness itself and that which comes to consciousness. This distinction in turn demands the union of the distinguished sides. An empty consciousness, a consciousness that is conscious of nothing, is a contradiction: consciousness must have a content.

Although consciousness and its content are to that extent essentially the same — since the passivity and activity of consciousness are always determined in likeness with that which, under this passivity and activity, is taken up into consciousness — they are nevertheless in reality so different that the scientific orientation depends on whether attention is directed especially either to the consciousness determined by the content, or to the objectivity of the content set over against consciousness.

Since the side of consciousness is everywhere, and must be, the predominant thing in philosophy, it is necessary to investigate the fundamental relation a) between consciousness and its content in general, b) between consciousness and the particular content, c) between consciousness and the absolute content. The main parts of the propaedeutic corresponding to this are then I) Theory of Knowledge, II) Theory of Science, and III) Doctrine of Ideas.

I. **Theory of Knowledge.**

§ 1.

Knowledge presupposes identity of consciousness and content; as the object is, so too must consciousness be: “the like is understood by the like.” Insofar as consciousness is sensing, the object is sensible; insofar as it is thinking, thinkable.

The essential difference between sensation and thought can be recognized by the difference between the Positive and the Negative. The object of sensation is, without regard to whether the sense is outer or inner, always something immediate, existing

in the moment, and to that extent a Positive: the Negative cannot be sensed. By suspending immediacy and positing negation, thought arises; one can think *about* the immediate, but its immediacy cannot be immediately thought. Insofar as thought begins with the immediate, the understanding shows itself as the intellectual sense. The sensing understanding is not a thinking, but the immediate presupposition of thinking, a thinking in possibility: actual thinking begins with the Negative. In its pure, qualitative opposition to the immediacy of sensation, the negation of thought is itself immediate. The immediately Positive and the immediately Negative bound one another; by this bounding they mutually determine each other. When it is asked what the Positive is, the answer must be given by means of a Negative, *omnis determinatio est negatio* [all determination is negation]; likewise, on the other hand, the Positive must be presupposed when negation and the Negative are to be determined.

As sensing, consciousness is essentially passive; here it holds that “all our knowing is a finding, receiving, and beholding (*scimus, quia accepimus* [we know because we have received])”; as thinking, consciousness is essentially self-active. As regards self-activity, all our knowing (Cf. Sibbern: *Om Erkjendelse og Granskning*) is “a doing and producing (*scimus, quia facimus* [we know because we make])”. The sensible is objectified in representation and image, the thinkable in insight and conception.

Sensation is not merely heterogeneous with thought: what can be sensed cannot be thought, and conversely, what can be thought cannot be sensed; but the individual senses are also mutually heterogeneous: what can be seen cannot be heard, what can be heard cannot be felt. Nevertheless these opposites too determine one another mutually. The audible can be determined by the visible (sound-figures), the sensible by the thinkable. In the passivity of the sensing consciousness there is an activity, in the thinking activity there is a passivity: the representation is obtained only insofar as consciousness itself forms it; thought determines only insofar as it is led, compelled, and determined by that which is thought; the sensing consciousness is thinking, the thinking sensing; the acting I is passive, the passive acting (*scientia et potentia in idem concidunt* [knowledge and power coincide in one and the same]).

Since it is now one and the same consciousness that is both passive and active, both sensing and thinking, and since one and the same object can have both sensible and thinkable properties, it follows from this that there must not only be an infinite reciprocal relation between sensation and thought, but that this reciprocal relation

must even be able to set itself up so differently that now it is thought that is absorbed into sensation, now conversely sensation that vanishes into thought.

Insofar as thought is absorbed into sensation in such a way that only the sensible in the object is granted validity, we get a one-sided sensuous tendency: *Sensualism*; the corresponding one-sidedness then consists in this, that sensation is grasped as the fleeting presupposition of thought, so that only the thinkable in the object, the negative essence, the pure concept, is acknowledged as the True: *Idealism*. The infinite reciprocal relation of sensation and thought will therefore first become evident when Sensualism and Idealism have each contributed their share to a closer determination of the truth in a double-sided *Factual Knowledge*.

A.

Sensualism.

§ 2.

The consciousness of the individual begins with an immediate unity of sensation and thought; but the advance of knowledge is conditioned by thought's being brought out on its own in its opposition to sensation. Now so long as the equilibrium between the two expressions of consciousness is preserved, so long as the agreement between the sensible and the intellectual is still maintained, individuality is happy, and its cultivation beautiful. But with growing abstraction, as the imagination really begins to work over and transform the elements of sensation, thought becomes restless, a tension sets in, and the equilibrium is lost. The lost equilibrium — whose mark is that now the intellectual, now the sensible, now an overwrought spirituality, now a bold naturalness, has gained the upper hand — will then be re-established, and the equilibrium thus re-established be lost again.

Corresponding transitions are seen in the history of philosophy.

Greek philosophy begins among the Ionians (Thales, Anaximander, Anaximenes) with a naïve cognition of the non-sensible in the sensible; thought seeks the principle of existence, and also divines that the principle must be “something unbounded,” yet finds it most nearly represented by a given sensible material: “Water,” “Air.” This

going-together of thought with sensuousness is not yet Sensualism. Thought is here so far from feeling the pressure of the senses' preponderance that it is precisely on the point of raising itself above the parcelling-out of the sensible; it is Oneness that thought has here grasped with youthful courage, as it first attempts a soaring flight toward the universal.

Only when thought has left immediacy behind, has consumed its content, and is weary of mirroring itself in itself, does Sensualism emerge. Sensualism is a conscious opposition, an express counterweight to the empty conceptual essence: only after Greek thought (Pythagoreans, Eleatics, Plato) had volatilized the material of sensation, held fast to the pure essence, and tried its hand at all manner of abstractions, does Aristotle assert experience, and the repressed Sensualism then properly comes forward in the theory of knowledge of the Stoics and Epicureans.

The stages which in this respect characterize the directions of the movement in Greek philosophy repeat themselves on an even larger scale in the universal history of philosophy.

In contrast to the Middle Ages, antiquity on the whole exhibits the beautiful equilibrium and natural unity of sensation and thought; but in the Middle Ages abstract spirituality commits such an overreach that a reaction thereby called forth has been able to lead the modern age to a denial of spirit whose extent, sharpness, and inner transparency is unknown among the ancients.

In the Middle Ages, when the natural is only something corruptible, the sensible only something sinful, the present only a deception, it is naturally impossible to know things as they are. Frightened by fancy and feeling, the thought of science enters the service of faith, and is cowed by the supernatural; the cowed thought finds only a formal freedom in the logical. Hence the emptiness, the subtlety, the word-splitting, and the scientific barrenness in which Scholasticism ends. Instead of grasping a reality, and nourishing itself on the working-up of a digestible material, reflection fed upon itself, and was tormented by hunger.

The efforts that scientific hunger called forth drew minds to nature in two mutually opposed directions, a speculative and an empirical; yet in such a way that something of the one can be found again in the other.

The speculative direction. Insofar as there can be any talk of a speculative Sensualism, we find such among the Italian philosophers: Telesio, Cardano, Bruno, and others.

With a decided predilection for the forms of ancient philosophy, these thinkers were inspired by the assurance of nature's identity with spirit, by the contemplation of that divine life which animates the world-organism, the living unity that works through all things.

That Aristotelian categories are applied to the interpretation of natural phenomena is not in and of itself Sensualism; the Scholastics did the same, in good understanding with the Hierarchy. The Sensualistic, on the contrary, comes to light in the boldness, frankness, and unconditionedness with which an omnipresent natural life displaces the Beyond, and sets itself in the place of the supernatural. The realm of reason with which Giordano Bruno deals (*De la causa, principio ed uno*) is at the same time a realm of the senses, and the wonders to which Vanini (*De admirandis Naturæ, reginæ Deæque mortalium, arcanis*) points are owed not to "the eternal King," but to "the Queen of mortals."

The empirical path. By sober research, by careful observation of the phenomena (Tycho Brahe) and corresponding calculations, that path was at last marked out which alone could lead to a true knowledge of nature. The law of falling bodies on earth (Galileo) and the planetary orbits in the heavens (Kepler) showed clearly enough that the path had been found. The investigators who thus understood not merely how to open the eye, but also how to arm the eye (the telescope), then, in the name of science, asserted the right of the senses, and set in force facts whose incontrovertibility must at length subdue the imaginings of the Middle Ages, and shake Scholastic metaphysics in its foundations.

Were such efforts to be grasped, judged, and presented in a theory of knowledge framed to the needs of natural science, then it would doubtless above all be a matter of impressing two chief demands: thought's liberation from abstract theories (from *anticipationes naturæ* [anticipations of nature]), and the leading-back of knowledge to true experience (*interpretatio naturæ* [interpretation of nature]). By working against formalism, emphasizing "Induction," and bringing the concept of natural law into harmony with the demand of physics, Francis Bacon of Verulam (†1626; *Novum organum scientiarum*), though without a natural-scientific initiation, has nonetheless, in inspired traits, pointed out a new foundation, and marked the transition to a new age.

§ 3.

Advancing Empiricism soon discovered the misinterpretation of the a priori that lay hidden in the doctrine of “innate ideas.” Against this misinterpretation J. Locke (†1704) directed his “philosophical essay on the human understanding” (*An essay concerning human understanding in four books*), whose content we may here consider under the following points of view: the origin of “ideas,” the essence of “ideas,” and the knowledge of things.

a) *The origin of “ideas.”* There are no innate ideas. Even if it could be shown that there were propositions about whose universality all must agree, this would still prove nothing, so long as the universal acceptance of these propositions could be explained in another way. But so far are such propositions from being found, that nations at different levels of culture are even at odds about the very most important practical principles. If one sets forth the commandment “Whatever you would have others do to you, do also to them,” then everyone who has not heard the commandment before, but thinks it over, will of course ask: why? But to ask for the grounding of a commandment that is innate is a contradiction.

As regards the theoretical propositions, such as “Whatever is, is; it is impossible that one and the same thing can at the same time both be and not be,” children, idiots, and the uneducated are wholly unacquainted with their meaning. But if such principles really were innate to the understanding, then all would have to be aware of them from earliest childhood, for to be in the understanding is the same as to be in knowledge.

Likewise it is an evasion when, in order to defend innate ideas, one appeals to the fact that these ideas immediately become evident to consciousness as soon as a person begins to use his reason. The relation is precisely the reverse: it is not universal but particular propositions, such as “Sweet is not bitter,” of which one first of all becomes conscious when one begins to think. That theoretical and practical principles should be innate is on the whole just as absurd as that arts and sciences should be innate. It is therefore impossible to be born with ideas; the child’s soul is like a clean sheet of paper, as empty as a *tabula rasa* [a blank tablet].

Whence, then, does the understanding get its ideas? From experience. But experience is of a twofold kind: insofar as it is made by an observation of external objects conducted through the senses, we call it sensation, *Sensation*; insofar as it

rests on observation of our own understanding's activity, we call it the inner sense, *Reflection*. Sensation and Reflection are to be regarded as "the only channels that convey knowledge to the understanding, the only windows through which the light of ideas falls into the dark chamber of the soul."

b) The essence of "ideas." The ideas coming from experience are partly simple, partly compound.

The simple ideas furnish the material of knowledge, are, so to speak, the letters in the alphabet of existence. In that these simple ideas press themselves upon consciousness from without, the soul is passive, and takes up the impressions of the surrounding world as the mirror takes up the object's image; if, on the other hand, they are obtained through Reflection — as the idea of thought, the idea of will — the soul is active. The simple ideas are partly such as come to the understanding through a single sense: colors through the eye, tones through the ear, impenetrability through the sense of touch; partly such as are conveyed to it through several senses, as when space and motion are apprehended both by touch and by sight.

By the connections and compositions of the simple ideas the soul can now know the objects of the surrounding world, "just as in language, by manifold compositions of letters, one can form syllables and words"; but, in that the compound, complex ideas are referred to three chief classes — modes, substances, and relations — it becomes above all the task to give an account of the concept of substance.

The nominal essence indicates the relation between genus and species; the real essence, on the other hand, denotes the inner peculiarity of the existing thing. In the simple ideas and modes the real and the nominal essence are one and the same; with respect to "substances" they are different. That a surface bounded by three straight lines is a triangle expresses the unity of the real and the nominal essence. It is otherwise with the part of matter that forms the ring on my finger; on the real peculiarity of the non-sensible parts depend the properties of its color, weight, softness, hardness, and so on. After the observed properties the name "Gold" is formed, an expression that denotes the nominal but not the real essence.

By sensation and reflection we experience how a number of simple ideas come together in a peculiar connection, presuppose on that account a necessary ground of connection, and refer this ground to a substratum subsisting on its own, namely *substance*; but is it now possible to know substance?

c) *The knowledge of things.* As the bearer of the qualities by which the simple ideas are produced in us, substance can be known only in this, that a distinction is made between the properties that necessarily belong to the thing in itself, and prove the agreement of the simple ideas with the objects (*qualitates primariæ* [primary qualities]), and such properties — color, tone, taste, smell, warmth, cold — as do indeed proceed from the things' action upon the soul, but are nevertheless wholly different from the objective essence of the things (*qualitates secundariæ* [secondary qualities]).

In this way Locke would determine the opposition between real truth (*real truth*) and mental truth (*mental truth*); but although substance is distinguished from all other complex ideas precisely in this, that it has its archetype outside us — one does not connect a sheep's bleating with the representation of a horse —, and precisely therein makes known its objective reality, it is nevertheless impossible for us to penetrate into substantiality itself. What we call the essential does not bear upon the individual. "It is doubtless necessary that I be what I am: God and nature have formed me thus; but I know nothing that is essential to me, the individual, as individual: an accident, an illness, can bring about incredible changes." So the essential bears, then, upon the species; but it can by no means be the real essence that determines the species, since the real essence is unknown to us. We are referred to a collection of sensible qualities that we can observe; but even when our observations are conducted with the greatest possible accuracy, our knowledge is just as far from corresponding to that inner constitution of the thing from which these qualities follow "as a peasant's representation of the clock in Strasbourg is far from corresponding to the inner, artful structure of that remarkable machine."

What Locke had begun, Condillac (†1780; *Essai sur l'origine des connaissances humaines*) continued. By letting the two sources of knowledge — sensation and reflection — flow together in such a way that the latter vanishes into the former, and by laying a reinforced emphasis on sensation, Condillac arrived at the result that the so-called inner sense is only a reflex of the outer, and that consequently all expressions of spirit represent only modified sense-impressions. Once Sensualism, thus carried through, has first shown that the True is only something sensible, it lies near to reverse the proposition and say: only the sensible is the True. Here is a materialism for which everything spiritual is a figment (La Mettrie).

§ 4.

From presuppositions by which a one-sided standpoint is expressly designated, consequences can be developed partly in a coordinate, partly in a superordinate direction. To consequences in a direction coordinate with the standpoint belongs everything that unfolds its character, and raises its significance to an ever wider compass. In this sense Condillac and La Mettrie have driven the consequences of Sensualism to the extreme.

To consequences in a direction superordinate to the standpoint belong those that lead out beyond the standpoint itself, and thus either establish or at least introduce a new standpoint. In this sense Hume's (†1776) Scepticism (*A treatise of human nature*) is to be grasped as a higher consequence of Lockean Sensualism.

To test the validity of concepts, one must — herein Hume agrees with Locke — hold to experience. Those concepts that can prove their origin from sense-impressions (*impressions*), or at least from the reflections corresponding to these, ought to be acknowledged; those that cannot, are of spurious descent, and ought to be rejected.

According to Locke, "the idea of cause and effect" arises when, in considering the constant change of things, we observe how certain things — be they substances or qualities — begin to exist through the law-governed activity of another. But what validity now has this observation? whence do we know that two things stand to one another as cause to effect? *A priori* we cannot know it; for *a priori* knowledge is analytic, but the knowledge of the causal concept must be synthetic, since the effect is really something other than the cause. Neither do we know it from experience; for experience shows us only that the one fact partly follows the other in time (succession), partly touches the other in space (contiguity); but no experience can teach us that the one essentially follows from the other, as the effect from its cause.

To assert the validity of substance Locke had already referred to habit; this indication Hume follows: "Our inferences from experience are grounded solely on habit; we are accustomed (*the mind is carried by habit*) to see the one thing follow partly along with, partly after the other, and then habituate ourselves thereby to the representation that the one must be an effect of the other. Out of the relation of contiguity and succession our imagination forms the causal relation: *cum hoc aut post hoc, ergo propter hoc* [with this or after this, therefore because of this]."

That the threads of the investigation here run together in the question of the causal concept as in their nodal point, shows the sureness with which the task has been posed.

Once the causal concept has fallen, then the categories — force, substance, necessity, truth — fall as of themselves; the bonds of essence that were to hold things together must now all burst.

As proceeding from a cooperation of sensation and reflection, the concept “force” might, in Locke’s opinion, well have its validity; but how is it now possible to know force? It might at first glance seem as though the relation between soul and body furnished, in this respect, a reliable presupposition. In that certain bodily organs involuntarily obey our will, spirit does indeed act as force, and we need therefore only to become conscious of our voluntariness in order to become conscious of an expression of force. This is, however, a deceptive reflection; partly we do not know the means by which spirit acts upon the body; partly it is not, after all, all the body’s organs whose activity is subject to the will. Were force to be an object of experience, then — what is impossible — cause and effect would also have to be an object of experience.

By presenting substance as an Unknown, Locke had already introduced the doubt about its existence. How are we now to know substance? Were the validity of substantiality to admit of proof, then the inner necessary connection of the properties that in substance are united into unity would have to admit of proof; but with causality the necessary connection (*this idea of a necessary connexion*) has fallen away. To refer to the difference between original and derived properties does not help. The derived properties are all subjective, and thus have only existence in our apprehension; as for the original ones, they let themselves at last be resolved into one single one, namely “Solidity,” but Solidity is an empty word, since we, when all the other properties are suspended, cannot possibly make the Solid an object of any experience.

In reliance on the “inner sense,” Locke had still held fast to “the idea of will,” and acknowledged the self. But how should it now be possible to assure oneself of the reality of the I? Our feelings, thoughts, and representations cannot possibly be regarded as expressions of force of an I, when the category expression-of-force has no validity; they cannot possibly be referred to an acting self, as to their true cause, when the causal concept has no truth.

The remnant of reality that still attached, in Locke, to the concept of truth (*real truth*) has thus vanished: truth is gone, probability has stepped into its place. The I, the self, is then in reality nothing other than a figment, which arises in this, that an aggregate of many rapidly successive representations is laid under a substratum.

That a philosophy which subordinates thought to sensation, even to such a degree that concepts must lose their universal validity, that the laws of things must vanish before the rules of the association of ideas, the ground of reason before habit, truth before probability, the self-knowledge of the I before a deluding figment, is in itself neither tenable nor satisfying, goes without saying; nevertheless Hume's Scepticism, in contrast to dogmatic Materialism, is a higher power, a spiritual force whose attack upon thought can transform itself into a defense of thought.

B.

Idealism.

§ 5.

If Sensualism can be recognized by this, that it examines the senses in order to judge thought, then Idealism can in turn be recognized by this, that it makes thought the judge of the senses; if Sensualism sets things above consciousness, then Idealism reverses the relation, and sets consciousness above things.

In that Descartes (†1650; *Principia philosophiæ*) begins with doubt, he wishes, like Bacon, to cast off prejudices, and free the mind from blind trust in authority; but, in that by doubting everything (*de omnibus dubitandum est* [everything is to be doubted]) he can also draw into doubt the validity of natural things and of experience as well, he goes further than Bacon.

Since doubt is a work of thought, thought is, in the midst of doubt, raised above doubt; everything can be doubted, but the validity of the doubting thought cannot be drawn into doubt; to doubt doubt is to confirm doubt.

Empiricism, which begins with an unconditional reliance on the immediate sense-impression, ends consistently in Scepticism; for Scepticism is an activity of spirit in

which thought, after having stared itself blind on the sensible, incessantly comes to itself as a stranger, tears itself down, and cannot know its own essence.

In the idealistic doubting of everything, thought begins precisely by finding itself, in that it begins by removing the foreign, and shaking off the appendage of tradition. This doubting is then so far from losing itself in Scepticism that the thinker rather wishes to assure himself of the validity of everything that withstands the testing of thought. The thinker then not only sees the unity of his thought with being (*cogito ergo sum* [I think, therefore I am]); but also gains certainty that consciousness is borne, penetrated, and filled by an absolute being, by the Unconditioned, whose image is in the soul, whose forms (*ideæ innatæ* [innate ideas]) shine with a clarity that can proceed only from the Perfect.

In Descartes, however, Idealism is still restricted in this, that over against the being that is in thought there stands a material being opposite thought; to suspend this substantial separation, and transform thought and being into attributes in one and the same substance, is an advance owed to Spinoza (†1677). Insofar as the question here is of the validity of thought and its identity with being, Spinozism is decidedly Idealism.

As regards the soul's a priori self-activity, Leibniz (†1716) went a step further. In his "Nouveaux essais" he works against Lockean Sensualism, and frees the assumption of "innate ideas" from the misunderstanding that could justify Locke's objections. To the Sensualists' *nihil est in intellectu, quod non antea fuerit in sensu* [nothing is in the intellect that was not previously in sense], there is then added a *nisi intellectus ipse* [except the intellect itself]. That "ideas" are innate must not be misinterpreted as though at birth they lay ready-made in the soul, since on the contrary they are in it only according to disposition. This view is in essential connection with the doctrine of monads; as monad the soul is originally self-active, can receive no impression from without, but on the contrary develops a world in its interior.

Whereas Locke compared the soul to an unwritten tablet, Leibniz compares it to "a block of marble in which the veins preform the shape of the statue." The principle of contradiction (*principium contradictionis*), the proposition of sufficient reason (*principium rationis sufficientis*), and the so-called *principium indiscernibilium* belong among "the innate ideas." How Idealism, by thinking over the relation between the senses and the sensed, and pointing out the contradictions that everywhere accompany the cognition of an external, can transform the whole outer world into

a world of spirit, has been shown in particular by G. Berkeley (†1754; *A treatise concerning the principles of human knowledge*; and *Three dialogues between Hylas and Philonous*). — Convinced that it is the assumption of matter's unconditional existence that especially favors Scepticism and Atheism, Berkeley reckons this assumption among the most meaningless and confused that can ever arise in a human head.

Sensible things are such as are immediately (*immediately*) apprehended by the senses, whereas the cause of things, which cannot be sensed, must expressly be thought, and, in being thought, be in consciousness. If I see that the one part of the sky is red, the other blue, then I infer that there must be a cause of this color-difference; but the cause cannot be said to be a sensible thing (*or perceived by the Sense of Seeing*). Nor can I, when I hear various tones, be said to hear the cause of the tones; just as little as I, when by feeling I experience that a thing is hot or heavy, can be said to feel the cause of the heat or the weight.

Sensible things are aggregates of sensible qualities (*Combination of sensible Qualities*); these sensible qualities are not outside the soul, but in the soul. As little as one can ascribe to the needle a sensation of the pain that its prick causes us, so little can one ascribe warmth to the fire. To have a high degree of warmth, to be hot, is to be in pain; were the fire really hot, then it would have to burn itself as it burns us, it would then have to be in pain; but that is absurd.

Not only of the derived properties, but of the original ones too, it holds that they exist only in our consciousness. Were properties such as extension, figure, weight, density, motion something outside consciousness, then one and the same thing would have to be both large (namely when seen near) and small (insofar as seen at a great distance), both smooth and rough, both straight and crooked, and the same motion both slow and fast; but that is absurd.

Just as absurd is the opinion that the named properties should have a substratum; the substratum on which extension and the other properties should depend would, in order to bring about extension, have to have an extension already beforehand, and so on to infinity.

The infinite divisibility of matter is generally assumed. There is then in every finite matter an infinite number of parts that the senses do not yet apprehend; but the more the eye is sharpened, the more parts it sees, and the larger each part becomes: could the eye be sharpened to infinity, each smallest part would become infinitely

large. This materialism contradicts itself; for it follows from this that every body is in itself infinitely extended, that is to say: without any shape or figure. It is therefore not an essence of matter, but consciousness itself, that forms all changes in the bodies of which the visible world (*the visible World*) consists: matter is an idea.

As for mathematical natural philosophy, mathematics forgets to give a sufficient account of its principles. Arithmetic reckons with units (*Unities*), but is not able to show what "the unit" really is; in geometry the infinite divisibility of finite extension plays a chief role; but finite extension is, as already shown, only an idea in the soul. Since it is now evident that no one is able to separate infinitely many parts either in a finite line or in a finite surface or in a finite body, everyone can, by making clear to himself the idea of extension, convince himself that the alleged infinity is not present. What in everyday speech we call things, we must then, according to the given explanations, call ideas. It is doubtless a hard saying, if for daily use we should eat ideas, drink ideas, and clothe ourselves in ideas; but usage is one thing, truth another. That things outside us are transformed into ideas in our own soul is not to be understood as though the sensible representations rested on our own arbitrariness. We can arbitrarily open and close our eyes, but the shapes and colors that show themselves to our opened eyes affect us with necessity.

Our sensible sensations and representations must doubtless be referred to a cause independent of us. But since cause and effect must be of like kind, and since a thing cannot possibly impart to us what it does not itself possess, it is just as absurd to assume that a thing that does not itself have sensation should be able to arouse sensation, as that a brain-impression should be able to become a representation. There exist therefore no bodies, but only spirits; it is not a material cause, but an almighty spirit, that through all sense-images, representations, sensations, and impressions lets the finite spirits impart themselves to one another.

By thus explaining the sensible and abolishing the material, this Idealism has transformed itself into a Spiritualism that annihilates nature, appeals to the supernatural, and culminates in Theology. *It is therefore plain, that nothing can be more evident to any one, that is capable of the least Reflexion, than the Existence of GOD, or a Spirit who is intimately present to our Minds, producing in them all that variety of Ideas or Sensations, which continually affect us, on whom we have an absolute and intire Dependence, in short, in whom we live and move, and have our Being.* (*Principles of hum. knowl.* London

1734. S. 166).

§ 6.

Among the many mutually crossing approaches to a theory of knowledge, Scepticism and Dogmatism meet as conflicting results. In Scepticism (Hume) thought is at variance with itself, and experience stands as an unexplained fact; in Dogmatism (Chr. Wolff) the unity of being with thought is grasped so immediately that the rigid schema of the understanding does indeed dazzle by its proofs, but lacks the grounding. Against these mighty one-sidednesses I. Kant (†1804; *Kritik der reinen Vernunft*) comes forward.

Descartes's Idealism, which makes the existence of objects in space unprovable and doubtful, Kant calls the problematic; Berkeley's, which presents the external existence of things as false and impossible, he calls the dogmatic. Dogmatic Idealism is unavoidable when space is to be regarded as a property belonging to things in and for themselves; for then every thing for which such a space should furnish the condition is an absurdity. The ground of this Idealism is suspended in "the transcendental Aesthetic." Problematic Idealism, on the other hand, is first refuted when proof is furnished that even our inner experience, not doubted by Descartes, is possible only under the presupposition of outer experience.

Although the Kantian critique of reason, as a systematic whole in itself, has by no means come into being out of finite regard for this or that view, it can nonetheless, if one will, be grasped in such a way that the two parts of "the transcendental Doctrine of Elements," namely the Aesthetic and the Logic, may chiefly be said to be directed against Scepticism, its third part, namely the Dialectic, against Dogmatism, and "the transcendental Doctrine of Method" against both. We therefore treat, in this connection, the Kantian theory of knowledge insofar as it characterizes itself, by its opposition to the named extremes, as a "*transcendental*" Idealism. Thus:

1) Against Scepticism.

That all our knowledge begins with experience is beyond all doubt; but from this it does not yet follow that knowledge in each and all has its source from experience. There are a priori sensible intuitions, a priori categories of the understanding, and a priori ideas of reason.

a) “The transcendental Aesthetic” deals with the pure sensible intuitions: space and time. Space is the form of the outer sense, time of the inner. If we abstract from the objects outside us, we get space back as the pure, universal form; if we abstract from the changes observed by the inner sense, empty time remains.

That intuitions of space and time are *a priori* can be shown partly from a metaphysical, partly from a transcendental point of view. The metaphysical consideration proves that the intuitions space and time cannot possibly be borrowed from experience, since one must precisely presuppose them in order to be able to make an experience; it proves further that space and time are not concepts but intuitions: the individual spaces are contained as parts in universal space, likewise the individual times in universal time, whereas the universal concepts, on the contrary, always comprise the particular ones as members under themselves, never as parts within themselves. The transcendental consideration proves the *a priori* origin of space and time in an indirect way, in that it shows that certain sciences, such as, for example, pure mathematics, would be impossible if there were no universal and necessary propositions about space and time; but universal and necessary propositions are never had from experience, they are had only *a priori*.

b) “The transcendental Analytic” treats the pure concepts of the understanding. The understanding is, negatively determined, a non-sensible faculty of cognition; outside intuition there is only a cognition by concepts. To intuition as receptivity corresponds the spontaneity of thought: intuitions rest on “affections,” concepts on “functions.” Intuition bears immediately on the object, the concept, on the contrary, mediately by means of another representation. The understanding judges; the judgment is a mediate cognition, a representation of a representation.

When we then abstract from the content of a judgment, and look only at the bare form of the understanding, we find that the function of thought lets itself be brought under four titles, each of which again contains three moments. In judgments there lie, namely — insofar as they are divided, according to Quantity, into universal, particular, and singular — the categories: totality, plurality, unity; insofar as they are divided, according to Quality, into affirmative, negative, (limiting), the categories: reality, negation, limitation; insofar as they are divided, according to Relation, into categorical, hypothetical, and disjunctive, the categories: substance and inherence, cause and effect, reciprocity; and finally, when they are distinguished, according to Modality,

into problematic, assertoric, and apodictic, the categories: possibility, actuality, and necessity.

That these twelve categories, from which the rest let themselves be derived, are not owed to experience, but are a priori concepts, is seen from this, that these too are just as necessary conditions for the possibility of an experience as the pure intuitions. In and for themselves the categories are empty forms that get their material conveyed only through intuition; only when the sense-content of intuition is absorbed into the form of the understanding does experience come about. Yet a difficulty meets us here. The objects, which are of sensible nature, are heterogeneous with the concepts: how then does it come about that a posteriori sense-objects can be subsumed under pure concepts of the understanding, and judgments *a priori* be formed therefrom? This happens in that a third, which on the one side is a priori, on the other side sensible, steps between, namely, the pure intuitions: space and time. In order in this respect to bring the concept into intimate union with intuition, the imagination forms “the transcendental schema.” Thus Quantity has for its universal schema the time-series or number, so that the pure concept of magnitude meets intuition, in that we in imagination produce units, and thereby go from unity through plurality to totality. Quality’s schema is the time-content: the Real fills a given time, and, when the content of reality is gone, negation is intuited in the emptiness of time. The schema of the categories of relation is the time-order, for substantiality is the persistence of the Real in time, the members of causality follow one another with necessity in time, and reciprocity is the regular coexistence of two or more expressions of substance in time. The schema of Modality is the time-comprehension: the schema of possibility signifies a representation’s agreement with the conditions of time in general, the schema of actuality an object’s existence in a given time, and that of necessity an object’s being at all times.

Since the objects we can observe all together fall in time, they must necessarily also fall under those schemata of time. This is proved in “the Analytic of Principles.” At the head of the analytic judgments stands the principle of contradiction; but the highest principle of the synthetic judgments lays down that every possible experience “must stand under the conditions that are necessary for bringing the manifold impressions of intuition to synthetic unity.” α) All phenomena are then, since they can be apprehended only under the forms of space and time, extensive magnitudes; “the Axioms of Intuition.”

β) All phenomena are, since every sensation can be strengthened and weakened in a definite degree, intensive magnitudes; “the Anticipations of Perception.” γ) All phenomena are, according to the schema of the categories of relation, subject to necessary determinations of time; the substantial remains in time; the accidental changes in time; the temporal change of accidents expresses a relation between cause and effect; “the Analogies of Experience.” δ) Finally all phenomena are, in agreement with the schema of the categories of modality, determined in such a way that what agrees with the formal conditions of experience is possible; what agrees with its material conditions, actual; what exhibits universally valid conditions of experience, necessary; — “the Postulates of Empirical Thought.”

c) “The transcendental Dialectic” investigates the ideas of reason.

Our cognition begins with the senses, passes thence over to the understanding, and ends in reason as in its highest. If the understanding is the faculty of bringing phenomena under a unity by means of rules, then reason is in turn the faculty of bringing the rules of the understanding to unity under principles. The understanding develops the conditioned, reason points out the unconditioned; the understanding furnishes the premises, reason draws out the conclusion. From the three logical inferences of reason, namely: the categorical, the hypothetical, and the disjunctive, the three speculative ideas of reason are derived: the idea of the soul as a thinking substance (the psychological idea), the idea of the world as the aggregate of all phenomena (the cosmological idea), the idea of God as the highest condition for the possibility of all (the theological idea).

That the ideas of reason are not abstracted from experiences is seen immediately from this, that in the whole world of experience no object can be pointed out that corresponds to any of those ideas. That these ideas nonetheless have validity, namely a priori validity, is understood from this, that they are the indispensable lodestars of the understanding, the regulative principles of cognition. We order our subjective faculties by means of the idea of the soul, the infinite series of conditions by means of the idea of the world, the ascending steps of causes by means of the idea of God.

2) Against Dogmatism.

The a priori is, according to Kant, solely in our consciousness, and has nothing to do with the essence of things. Already “the transcendental Aesthetic” warns us against

regarding space and time as being outside us. Were one to set up the proposition that things are in space and time, then it would be a delusion; not in themselves, but only for us are things in space and time, and that in such a way that the phenomena for the outer sense are in space as well as in time, the phenomena for the inner sense only in time. Just as emphatically “the transcendental Analytic” impresses that the categories of the understanding have universal validity only insofar as they hold for us, in that we apply them to things as objects of experience, but not to things in and for themselves. In that we see objects through subjective forms, and apprehend them in subjective concepts, we are like those who see the colors of things through painted spectacles.

The discord between thinking and being that Kant thus brought about by impressing from first to last the separation of the thing as object of experience and the thing in itself (*das Ding an sich* [the thing in itself]) — a separation that is further illuminated by the way in which, with respect to *Phænomena* and *Noumena*, he demonstrates “the Amphiboly of the Concepts of Reflection” — is well suited to give uncritical Dogmatism a fundamental blow; this blow becomes especially palpable in the critique that the ideas of reason must undergo in “the transcendental Dialectic.”

The Dialectic, “*eine Logik des Scheins* [a logic of illusion],” is to clear up the illusions in which reason, by applying the categories of the understanding to the unconditioned, must necessarily entangle itself. This “transcendental illusion,” this optical self-deception of reason, shows itself in the different ideas of reason from a different side.

a) The Paralogisms of pure reason. Rational psychology makes the soul into “a soul-thing with the attribute of incorporeality (immateriality); into a single substance with the attribute of incorruptibility (incorruptibility); into a numerically identical intellectual substance with the predicate of personality; into a spaceless, thinking being with the predicate of immortality.” These propositions of rational psychology are sheer fabrications. At the ground of the four paralogisms belonging here can lie nothing other than the simple, in and for itself empty representation: “I think”; but this “I think” is neither intuition nor concept, it is only an act of consciousness that accompanies, bears, and connects all my representations and concepts. That I who think can always be regarded as subject, a something that does not merely, as a predicate, hang upon thought, is an apodictic, indeed even identical, proposition; but

it does not signify that I as object am a being or substance subsisting for myself. The proofs of immortality rest on deceptive inferences. "I can indeed separate my pure thinking ideally from my body; but from this it by no means follows that my thinking can also persist really, after being separated from the body."

b) The Antinomies of pure reason. To determine the chief sides from which the idea of the world shows itself, one is guided by the categories given in the table. In respect of the Quantitative, there is then here something to lay down about the totality of the world's times and spaces, and as regards Quality, something about the divisibility of matter; in respect of Relation the causality of the effects given in the world must be pointed out, and, as regards Modality, the conditions of the contingent are pursued until at last they are absorbed into the necessity of the whole.

At each of these points, assertion stands against assertion (antithesis against thesis) with proof of equal force. Against the thesis: "The world has a beginning in time and is enclosed within the limits of space," holds the antithesis: "The world is infinite in respect both of time and of space"; against the thesis: "Every composite substance in the world consists of simple parts," the antithesis: "No composite thing in the world consists of simple parts"; against the thesis: "Besides causality according to natural laws there holds in the world a causality of freedom," the antithesis: "there is no freedom, but everything in the world happens solely according to natural laws"; against the thesis: "To the world belongs something that, either as a part of it or as its cause, is a simply necessary being," the antithesis: "There exists no simply necessary being as world-cause, neither in the world nor outside it."

These cosmological antinomies then show us the dialectical struggle, the self-contradiction, in which reason entangles itself when by the idea of the world it wishes to think the objective essence of the world.

c) The Ideal of pure reason. "The category" is empty, but can nonetheless find its object in experience; "the idea," which has no object of experience at all, is thus more remote from objective reality than the category. "But still more than the idea, that seems to be removed from objective reality which I call the Ideal, and by which I understand the idea, not merely *in concreto* [in the concrete], but *in individuo* [in the individual], i.e. as a single thing determinable and determined by the idea alone." After having shown how reason determines its Ideal in different ways, namely as primal being (*ens originarium* [original being]), insofar as it has it in itself, as the

highest being (*ens summum* [highest being]), insofar as it has it above itself, and as the being of beings (*ens entium* [being of beings]), insofar as everything is conditioned by it, Kant passes over to an attack on the “speculative reason” of Theology, and in his critique of the dogmatic proofs of the existence of God drives the separation of thinking and being to the extreme.

Among the three proofs, the ontological, the cosmological, and the physico-theological, the emphasis must doubtless fall especially on the first. “On the impossibility of an ontological proof of the existence of God” the following should here be brought out. The proof is in brief this: “The most real being is possible; among the many realities belongs also existence; it therefore lies in the concept of the most real being that this being itself must necessarily exist.” One clarifies this by examples and dazzles by examples. When I suspend the predicate in an identical judgment, and keep the subject, then quite rightly a contradiction arises, and therefore one says that the predicate belongs to the subject with necessity. But if I suspend the subject together with the predicate, then no contradiction arises; “*denn es ist nichts mehr, welchem widersprochen werden könnte* [for there is then nothing more that could be contradicted].” To posit a triangle and yet suspend its three angles is a contradiction; but to suspend a triangle together with its three angles is no contradiction. So it is with the absolutely necessary being. God is almighty; almightiness cannot be suspended when you posit God; but when you say: God is not, there is neither almightiness nor any other predicate. Being is, as copula in the judgment, no real predicate. The actual contains in the concept no more than the merely possible. A hundred actual dollars contain not the least more than a hundred possible ones; for in the opposite case the concept would not be adequate. A concept loses not a single one of its properties because its object does not exist, and gains not a single property from its object’s being found existing.

“On the so famous ontological (Cartesian) proof, then, all trouble and labor is wasted; a person would, by mere ideas, become no richer in insight than a merchant in fortune by adding a few zeros to his cash balance.” From thinking to existence there is no speculative transition.

3) Transcendental Idealism.

Since Kant expressly understands by “a transcendental cognition” one that “does

not properly have to do with objects, but only with our cognition of objects, insofar as this is to be possible *a priori*,” it is already seen from this why he himself calls his Criticism a transcendental philosophy. This philosophy will now doubtless in its way assert the validity of the world of experience; but so would Berkeley’s Spiritualism too, in its way. The manner in which the moment of subjectivity becomes the overreaching thing in all our cognition characterizes the Kantian transcendental philosophy as Idealism.

In what relation this transcendental Idealism has expressly set itself both to Scepticism and to Dogmatism is expressed especially in “the Doctrine of Method,” which treats the pure reason’s *Discipline, Canon, Architectonic, and History*.

“Were one,” it says, “to ask the cold-blooded David Hume, properly created for the equilibrium of judgment, what then moved him, by scruples that only a laborious brooding could devise, to undermine the imagination so comforting and useful to men, that their rational insight suffices when the concept of the highest being is to be determined, then he would probably answer: Nothing, but the intention of bringing reason further in self-knowledge, and therewith a certain displeasure at the compulsion one would impose on reason, in that one prides oneself on it, and yet hinders it from making a candid confession about the weaknesses that it can discover in itself only by serious examination.”

Unsatisfying as Scepticism is in and for itself, so justified is it over against an uncritical Dogmatism that does not know what the understanding is capable of, and has not tested the limits of the possibility of cognition according to principles. “*Und so führt der Skeptiker, der Zuchtmeister des dogmatischen Vernünftlers, auf eine gesunde Kritik des Verstandes, ja der Vernunft selbst* [And thus the sceptic, the disciplinarian of the dogmatic rationalizer, leads to a sound critique of the understanding, indeed of reason itself].”

In pure cognition as science, reason must on the contrary always proceed dogmatically, i.e. conduct strict proofs from secure principles *a priori*; but to conduct strict proofs from secure principles *a priori* is not the same as (*Wolffs methodus mathematica* [Wolff’s mathematical method]) to imitate mathematics. Mathematics’s thoroughness rests on definitions, axioms, and demonstrations; philosophy’s on explications, deductions, and acroamatic proofs. It does not befit philosophy “*mit einem dogmatischen Gange zu strotzen und sich mit den Titeln und Bändern der Mathematik auszuschnücken*,

in deren Orden sie doch nicht gehört [to strut with a dogmatic gait and adorn itself with the titles and ribbons of mathematics, in whose order it does not after all belong].” The apodictic propositions (be they now provable, or else immediately certain) divide into Dogmata and Mathemata. A direct synthetic proposition from concepts is a *Dogma*, whereas a similar proposition by construction of concepts is a *Mathema*. Since there is now in the speculative use of pure reason not a single Dogma, every dogmatic method too — be it now borrowed from the mathematician, or in another way become a peculiar mannerism — is in itself unfitting. “*Denn sie verbirgt nur die Fehler und Irrthümer, und täuscht die Philosophie, deren eigentliche Absicht ist, alle Schritte der Vernunft in ihrem klärsten Lichte sehen zu lassen* [For it only conceals the faults and errors, and deceives philosophy, whose proper aim is to let all the steps of reason be seen in their clearest light].”

§ 7.

“In every experience two factors are set against each other: the I and the thing, the intelligence and its object. If one abstracts from the I, then one gets a thing *in and for itself*, and must then grasp the representation as a product of the object; if one abstracts from the object, one gets an I *in and for itself*, and must then hold to the self-active consciousness that produces its own content. The former is *Dogmatism*, the latter *Idealism*. These two are mutually incompatible; a third is not given: here one must choose. To judge between Dogmatism and Idealism, let one note the following: the I occurs in consciousness, the thing in itself is a fiction of consciousness. In that Dogmatism is to explain the origin of the representation from the thing in itself, it performs the dubious feat of going behind consciousness, and invoking being; but being effects only being, not consciousness; therefore only that standpoint can be true which is chosen, not in being, but in consciousness, not in the object, but in the I itself.” In this spirit J. G. Fichte (†1814) has drawn a consequence of Kantianism and (*Erste Einleitung* 1797) introduced his “*Wissenschaftslehre* [Doctrine of Science].”

While Fichte dissolves “the thing in itself,” holds fast to the I, and carries through *subjective Idealism*, the Kantian doctrine is illuminated from an opposite side, in that J. F. Herbart (Metaphysik) fixes attention on “the thing,” grasps even the I as a thing for itself, a “real with many properties” (Psychologie), and in his way develops a *critical Realism*. “The material of experience is the foundation of philosophy; but

as a mere given it still stands outside philosophy. What then is philosophy's first step, the act with which it begins? Experience has forced upon us concepts that do not let themselves be thought. They are doubtless thought in a way by everyday understanding; but it is an obscure and confused thinking that does not see through the conflicting distinguishing marks. The sharpened thinking, metaphysics's logical analysis, discovers in the concepts of experience (space, time, motion, and so on) contradictions, conflicting, mutually excluding marks. What is to be done here? These concepts cannot be thrown away; they are given to us, and we have only the given to hold to. Neither can they be kept, since they are unthinkable, in themselves impossible. The only way out is: we must rework them. *The reworking of the concepts of experience is the task of Speculation.*"

Returning, after this indication of the Herbartian direction, to Fichte, we must, in order to understand subjective Idealism, well imprint on ourselves the highest principles of "the Doctrine of Science"; for in these principles "the Doctrine of Science" is theory of knowledge.

α) *Thesis*. If one grants that the proposition $A = A$ is self-evident, then one grants at the same time consciousness's power to posit something. One posits not that A is, but only that, if A is, then it is A . If we now, instead of looking at what is posited, look at the positing one, or rather: at the identity of the positing one and what is posited, then the proposition $A = A$ transforms itself into $I = I$. Whereas instead of $A = A$ we could not say that A is, we can now, instead of $I = I$, say: I am. This self-grounded is the ground of all action in the human spirit, and thus the original distinguishing mark of the activity of consciousness in and for itself.

β) *Antithesis*. The proposition $\text{Non-}A \text{ not} = A$ is, as a free primal act, according to its form likewise unconditioned, but on the contrary, according to its content, its matter, conditioned, because the antithesis presupposes the thesis, because when a $\text{Non-}A$ is to be posited, an A must already beforehand have been posited. By the first principle, $A = A$, the form of the primal act was a positing; by the second it is a counter-positing. What $\text{Non-}A$ is, is not yet known; only this is known, that $\text{Non-}A$ is the opposite of A . But now A is posited by the I ; here only an I is presupposed: what is opposed to the I is thus $\text{Not-}I$.

γ) *Synthesis*. The third principle is, since it is determined by the two foregoing, almost entirely capable of receiving a proof. The task for the act expressed by the

third principle is given with the two foregoing, but not the task's solution. On the one side the I is completely suspended by the Not-I: the I cannot be posited insofar as the Not-I is posited. On the other side the Not-I, posited by the I, is itself an I, posited in consciousness. The I is thus not suspended by the Not-I; it is suspended, and yet nevertheless not suspended. This is the contradiction. For the solution of this contradiction we must seek to find an *X* in which the mutually excluding principles can be united. But how can being and not-being, reality and negation, be united with one another without annihilating one another? Answer: by mutual limitation! The *X* asked for is thus a barrier, and self-limitation the primal act of the I. Insofar as the essence of I-hood, the pure I, is grasped as a principle that develops itself in the empirical I's reciprocal action with a world of barriers, subjectivity has to such a degree gained the preponderance that herein is contained a natural summons to lay an emphasis of ideality on the objective too.

In his work "*Von der Weltseele*" F. W. J. Schelling (†1854) has, still with Fichte's presuppositions, sought to show how the concept of matter necessarily proceeds from the nature and intuition of the human spirit. "The mind (*das Gemüth* [the mind]) is the unity of an unlimited and a limiting force. Absence of barriers would, as well as absolute limitedness, make consciousness impossible. Only in that the force that strives out into the unlimited is limited by the opposite, while the limited one is conversely freed from its barriers, is feeling, observation, cognition thinkable. A corresponding thing is seen in nature. Not matter as matter, but the forces whose unity matter constitutes, is here the first. Matter is the ever-becoming product of attraction and repulsion. Matter is what it is by force; but force is the immaterial in matter, force in nature is that which can be compared with spirit. Nature and spirit must originally be united in a higher identity. The spirit's organ for the apprehension of nature is intuition; it is intuition that takes into possession the space bounded and filled by attractive and repulsive forces."

With these thoughts Schelling has already trodden the objective ground of natural philosophy. "All knowledge," it says (*System des transcendentalen Idealismus*), "rests on the agreement of a subject with an object. The aggregate of the merely objective is nature; the aggregate of the merely subjective is the I or the intelligence. For the union of both sides two ways are possible; either one makes nature the first, and asks how the intelligent comes to it, i.e. one strives to interpret it in pure forms of

thought — natural philosophy; or one makes the subject the first, and asks how the objects proceed from the subject — transcendental philosophy. All philosophy must aim either at making an intelligence out of nature, or a nature out of the intelligence. Just as transcendental philosophy has to subordinate the real to the ideal, so natural philosophy must seek to explain the ideal from the real. But both are only the two poles of one and the same knowledge, which mutually seek each other; therefore one must also, when one starts from the one pole, necessarily come to the other.”

“All the parts of philosophy must be presented in one continuity, and the whole of philosophy must be presented as what it is, namely as the continuous history of consciousness, for which what is deposited in experience serves only, so to speak, as monument and as document.”

Schelling’s transcendental Idealism can now be regarded as forerunner of Hegel’s “*Phänomenologie des Geistes* [Phenomenology of Spirit]”; what Schelling promised, Hegel (†1831) fulfilled.

“Consciousness has in general, according as the object shows itself, three stages. The object is namely either an object standing over against the I, or it is the I itself, or something objective that nevertheless belongs just as much to the I, to thought.” The three stages are thus: Consciousness, Self-consciousness, Reason.

a) *Consciousness*. At the first stage consciousness is sensing, observing, understanding.

α) *The sensing consciousness is immediate certainty of an external object*. The expression for such an object’s universality is, *that it is, and is this, now according to time, here according to space*. The object is thereby wholly different from all other objects, and for itself sufficiently determined. But this Now is no more, another has stepped into its place, which however is also a Now: the Now is, in its vanishing, something abiding, namely the universal. This Here, which I mean and point out, has a right, a left, an above, a below, a behind, a before, to infinity. This Here is thus not the single, immediately determined, but: now here, now here, now here! an aggregate of Many, and in these Many nevertheless the One, thus the universal.

β) The universal, insofar as it still hovers between sensuousness and reflection, is object for the *observing* consciousness; *here is the thing with its properties*. The sensible properties are at once immediately each for itself, and yet in relation to one another, belong to the thing as this single thing, and are yet so universal that they, separated

from one another, can be found in manifold other things. Thing and properties change, and yet have in this change their subsistence: the changeable is an outer, the subsisting an inner.

γ) Insofar as the object is an inner with its outer, a force in its expression, it is there for the *understanding* consciousness. The understanding sees in the changing outer of things a series of phenomena whose inner consists in a system of abiding relations: the laws. Things are under the laws, and the laws are the laws of things; things and laws are two sides of the same: the True, the essence of things, is thought, the concept.

b) *Self-consciousness*. In that consciousness turns toward thought as thought, toward essence as essence, toward the inner as inner, it gets its own reflection and thereby itself as object. Self-consciousness has in its formation or movement three stages: desire, the relation of ruling and serving, universal self-consciousness.

α) *In desire self-consciousness relates to itself as single, solely existing self-consciousness*; it is a selfless object that is to satisfy the craving; but, in that the object is consumed, and satisfaction sets in, consciousness has in its self-feeling understood that it, as subjective, must at the same time essentially be objective.

β) The concept of self-consciousness as a subject that is at the same time an object finds its complete expression in this, that self-consciousness sees in its object another self-consciousness. In that the two self-consciousnesses set over against each other are each according to their concept independent, the tension arises between being master and being servant. The servant is selfless, insofar as his self is a moment in the master's consciousness; but the master too is selfless, insofar as his being master rests on his having a servant; he is master only by means of the servant. With this reciprocity self-consciousness makes the transition to articulated universal will, to positive freedom.

γ) *In universal self-consciousness the individual knows himself* as serving the will of the many, in that he essentially serves himself. Here there is not a sum of wills, but the many wills are moments in the universal will. This self-consciousness is the foundation for all virtues: those of love, honor, friendship, bravery, sacrifice, distinction, and so on.

c) *Reason*. Reason is the highest union of consciousness and self-consciousness in knowledge of the object and knowledge of itself. Reason is the certainty that its

determinations are just as much objective, determinations of the essence of things, as that they are the subject's own thoughts. Fichte's "I" and Herbart's "real" have hereby become moments; reason's knowledge is not merely subjective certainty, but truth; but truth is unity of certainty and objectivity, of knowledge and being.

"The goal, absolute knowledge, or the spirit knowing itself as spirit, has for its path the recollection of the spirits as they are in themselves, and accomplish the organization of their realm. Their preservation, in respect of their free existence appearing in the form of contingency, is history, but, in respect of their conceived organization, the phenomenal knowledge's science; both together, the conceived history, form the recollection and the Golgotha of absolute spirit, the actuality, truth, and certainty of its throne, without which it would be the lifeless, solitary; only out of the chalice of this realm of spirits foams up for it its infinity."

C.

Factual Knowledge.

§ 8.

If we now look back on the theories of knowledge dealt with here, the sensualistic as well as the idealistic, then it is quite striking to us that not a single one among them all satisfies "sound understanding" in its consequences, while they nevertheless in their cooperation — insofar as each on its own most forcefully asserts its side of the matter — as it were unite in giving "sound human understanding" its right.

From this it cannot now be concluded that "sound understanding" is wiser than philosophy, but only that it, with the sureness of instinct, holds fast to the members whose mutual relation philosophy wishes to explain. When philosophy, then, instead of clearing up the relation of the members, explains away or bungles an essential member, "sound understanding" senses mischief, and protests.

"Sound understanding" readily grants that one cannot become conscious of a surrounding world without taking up its impressions and corresponding images into one's

circle of representation; but it maintains at the same time that the world we become conscious of must, as truly as it is a real world, have a real existence independent of our consciousness. The idealist may now exert himself to prove the indissoluble unity of world and consciousness as much as he will: “sound understanding” pro- tests. That our sensations and representations proceed from immediate sense-impressions, the understanding will readily grant; but when the sceptic draws from this the consequence that cause and effect exist only in our imagination, that the laws are not laws, “sound understanding” stands still, and begins to protest.

A philosophy that is in open contradiction with “sound understanding” is false; a philosophy that immediately conforms to “sound understanding” is superficial. Philosophy will not arbitrarily contradict the natural understanding, but only show that the so-called sound sense, without having any inkling of it, incessantly contradicts itself.

“Sound understanding” demands that I should immediately become conscious that there is something outside my consciousness; but in becoming conscious of something, it is of course in my consciousness; insofar as it is outside, I cannot become conscious of it: thus I am to become conscious that what is in my consciousness is not there; what a contradiction! When the sun shines, and the trees are mirrored in the stream, “sound understanding” means that the light is brightness, whether or not there is anyone who sees it, and that the trees really are mirrored in the stream, whether or not this mirroring is reflected in a seeing eye. But the question is how one is to convince oneself of this. By sight? That is a contradiction, since here it is a matter of the phenomenon, not insofar as it is seen, but insofar as it is not seen. By thought? That is a contradiction; the phenomenon does not let itself, as phenomenon, be thought, for in that case the one born blind could also think it. By the testimony of others? An empty evasion; since here it must go for the one as for the other. Only for the seeing one is light brightness; only in that the brook and the tree are mirrored in the eye can the tree itself be mirrored in the brook.

It falls naturally that one thinks back to the time when he did not yet exist; and just as naturally that he thinks of the moment when he will again have ceased to be. Nonetheless there is here in both a strange contradiction. That consciousness can be conscious of its own existence is evident; but how can the existing consciousness become conscious of itself as non-existing? In reality? No, that is a contradiction, for

when consciousness really is, its non-being is really excluded; and, when consciousness is not, it cannot become conscious of anything. In fantasy? But when consciousness sees its non-being only as a fantasy, how can it then convince itself that the real either has corresponded or will correspond to this fantasy? “Sound understanding” may readily advance with its explanation, but the matter is that the explanation contains just as many contradictions as that which it is to explain.

What is truth? Agreement between what consciousness takes to be real, and this real itself. And untruth? Naturally, conflict between what is posited in consciousness and what is in reality. If now all reality were at once gone, and the empty consciousness with its fantasies and thoughts stood alone behind, it would be able to carry on the most arbitrary play with figments. That truth is gone when the real world-content is gone, sound understanding will surely readily grant; but the real as merely real is, when consciousness itself is gone, not the True either. A stone, a star, any real object whatever, is in and for itself neither something true nor something untrue. In order that the difference between true and untrue can arise, a judgment must be passed on the real object; but in order that a judgment can be passed, there must absolutely be a judging consciousness. But if consciousness is an indispensable requisite for truth, the real for itself cannot possibly be the True; how then can “sound understanding” suppose it can think a time when there were in *truth* real atoms, real elements, a real earth, but yet no, no consciousness at all? Real objects that could not possibly be objectified; what a contradiction!

The outer, real world is, in its pure unconscious reality, a nothing; if anyone will maintain the opposite, he must maintain it by force of consciousness, and thus contradict himself.

What is an Eternal? When a universally valid proposition is known, it does not look at all as though the True first came into being in the moment of cognition; the True shows itself on the contrary to consciousness as that which has always been. That every thing is what it is; that two magnitudes, each of which for itself is equal to one and the same third, are mutually equal; that a ground must necessarily have its consequence, an effect its cause — all these so-called “eternal truths” present themselves to us as those that have been prior to all finite knowledge; and yet they are in themselves nothing when one takes knowledge away: how can the consciousness that comes to be in time become conscious of the non-becoming, the eternal?

What does contradiction mean? An existing nothing is a contradiction; a round square is a contradiction; a flying horse is also a contradiction. Nonetheless these contradictions are of different kinds. A flying horse one can represent to oneself, but not think. The unthinkable lies not straightforwardly in the logical, that the predicate “flying” contradicts the concept “horse,” for this concept could of course be extended; but in this, that the whole concept thus extended stands in conflict with the conditions and laws of experience. It is otherwise with the round square; here the contradiction is seen immediately in the representation itself. To represent the square as round is impossible: to think a round square is, on the other hand, possible, insofar as it is possible to think the predicate’s *conflict* with the subject. The contradiction is thus not merely that the predicate’s conflict with the subject makes the judgment meaningless; but the contradiction is in a higher sense this, that the contradiction itself, this unthinkable, nevertheless lets itself be thought. This becomes especially clear when the task is to think the existing nothing. It is indeed nothing, but in that consciousness posits this negation, nothing has precisely existence in thought, thus a thought existence.

By pondering a little on these many contradictions “sound understanding” will best learn to understand that the eager strife the philosophers carry on about “the principle of contradiction” is more than a strife about words.

Herbart and Hegel are both agreed that the concepts of experience contain a contradiction; they are furthermore agreed that the contradiction is to be suspended; but about the relation of the contradiction to the essence of things, indeed about the very essence of the contradiction, they are in the highest degree at odds. In that Herbart holds fast to the principle of contradiction, he sees in the contradiction an error that one is to avoid, a delusion that one is to remove. If the concepts of experience contain contradictions, then the fault lies not in the objective nature of things, but in the knowing subjectivity. The thinking consciousness must therefore, by reworking the concepts of experience and doing away with the contradiction, see to remove the false apprehension. Zealous against the contradiction, Herbart even accuses the Hegelian philosophy of Empiricism, since it “takes up the concepts from experience unchanged, and, despite the insight into their contradicting nature, precisely because they are empirically given, regards them as justified, indeed even for their sake remodels logic.” Hegel, on the contrary, with his “*Alles ist ein Widerspruch*

[Everything is a contradiction],” lays the contradiction into the essence of things, so that a being in finitude is necessarily a being in contradiction (solution of the Kantian antinomies). On contradiction rests not only the pain of life, but also the drive that drives existence forward: contradiction calls forth motion, and motion is in turn the solution of the contradiction. As little as existence can exclude contradiction, so little can thought exclude it: the pulse of thought is dialectic, the contradiction the nerve of dialectic.

§ 9.

How intimately contradiction is united with thought is seen in the logical judgments. The word “leaf” can, as a word in language, point to a sensibly given in its immediate singularity, but also, as a conceptual expression, designate something universal. An exclamation such as “this leaf!” is in linguistic form imperfect, and gives a definite meaning only when someone at the same time points to a real leaf. If the eye is now fixed on “this leaf,” thought can be united with the sensible, in that a judgment is uttered: “this leaf” is green. But “this leaf” is not green, for it is yellow; “this leaf” is thus green and yet not green; “this leaf” is both green and yellow; here is a swarm of contradictions. In the first place the predicate “green” is, in relation to “this leaf,” something universal, whereas “this leaf” itself is something single. The judgment thus states that the single is a universal; but the single is nevertheless not a universal, but a single: here is a contradiction. What then is “this”? “This” is a leaf, “this” is a table, “this” is a horse; “this” is thus not “this,” but something other than “this”: sheer contradictions.

Through the history of philosophy runs a series of controversies that all point to the contradiction present in the matter. In Stilpo’s utterance: “the cabbage that is offered for sale here, is not; for cabbage existed already 1000 years ago,” lies the meaning that the individual in its sensible immediacy cannot be thought. The Scholastics’ controversy about the relation between *res* [things] and *universalia* [universals] points to the same contradiction, which also, as regards the more recent philosophy, repeats itself in the opposition between Sensualism and Idealism.

Locke and Hume are both attentive to the delusion that can arise in this, that the universal is held fast for itself in the word; and, since the objective Idealism had advanced so far that Hegel’s Dialectic was assumed to have transformed the sensible

into a moment vanishing in the concept, Feuerbach (*Philosophie der Zukunft*) came forward with a critique that demanded unconditional Sensualism.

For the understanding of Idealism's one-sidedness we bring out in this connection the relation between the sensible and the thinkable, between the individual and the universal, between the real and the essential.

a) The sensible and the thinkable.

That the sensible is an other of thought, a barrier of reflection, Idealism must of course grant; but, in that reflection shows that thought precisely needs this barrier, the otherness, the barrier itself, is nevertheless thought after all, and the sensible, the Negative, which thought thinks in thinking itself, is thereby melted down into a moment in the concept. This ambiguity can for a part be veiled in the pure, objective Logic, where "matter," this illogical, nonetheless by the help of Dialectic at last becomes just as logical as "form"; but in "natural philosophy," where reflection strikes against the fact, the error is laid bare.

"By the old, general thought that every body consists of four elements, or the more recent Paracelsian one that it consists of mercury or fluidity, sulphur or oil, and salt (Jacob Böhme called it the great Trinity), and by many other thoughts of this kind, the refutation has in the first place been made easy, in that under those names one would understand the individual, empirical materials that are most nearly designated by these names. But it is not to be mistaken that they should much more essentially contain and express the conceptual determinations. One must therefore much rather admire *die Gewaltsamkeit, mit welcher der Gedanke, der noch nicht frei war, in solchen sinnlichen besondern Existenzen nur seine eigene Bestimmung und die allgemeine Bedeutung erkannte und festhielt; er darf darum auch nicht auf experimentirende Weise widerlegt werden*: [the violence with which thought, which was not yet free, recognized and held fast, in such sensible particular existences, only its own determination and the universal meaning; it may therefore also not be refuted in an experimental way:]" (Hegel, *Naturphil.* S. 271). This apprehension is quite in the spirit of Idealism, and that Hegel too will not have it restricted to an assessment of that thought "*der noch nicht frei war* [that was not yet free]," but even expressly applied in favor of his own interpretation of the phenomena, is evident from the way in which he himself, e.g., introduces the "lunar" and the "cometary" principle; for precisely in this "lunar" and this "cometary" is seen "*die Gewaltsamkeit, mit welcher der Gedanke in solchen*

sinnlichen besondern Existenzen [the violence with which thought, in such sensible particular existences],” recognizing and holding fast only “*seine eigene Bestimmung und die allgemeine Bedeutung* [its own determination and the universal meaning].”

The fundamental error (πρῶτον ψεῦδος) lies in the misprision of the sensible. The sensible is by no means suspended because it is grasped as thought’s other; the element of sensuousness is more than a logical negation, it is for thought the real barrier, the in-itself-unthinkable, which is therefore to be grasped and determined in relation to the immediate sense.

“According to the moment of its ‘being-in-itself,’ matter would be the single point; according to the moment of its ‘being-outside-itself,’ it would most nearly be a multitude of mutually excluding atoms. But, in that these, by excluding one another, relate to one another, the atom has no reality, and the atomistic as well as absolute continuity or infinite divisibility is only according to possibility present in it.” In this Hegelian apprehension it does indeed at first glance look as though thought penetrated matter. By the logical opposition of “*Insichseyn* [being-in-itself]” and “*Außersichseyn* [being-outside-itself]” and its dialectical treatment one does indeed escape the difficulties that Atomism, this stumbling-stone of Speculation, otherwise brings with it when it is to be clarified in a way satisfying to those versed in the matter; but the difficulty is only got around, not suspended: in that the element of sensuousness asserts its right, the relation between matter’s discretion and continuity takes up an unthinkable, a for reflection impenetrable; for the sensible is thought’s barrier.

To convince ourselves from yet another side how unthinkable the sensible is, one need only cast a glance at the Herbartian view. With objective Idealism, Herbart’s Intellectual-Realism namely has this in common, that it, though in its way, will make the sensible thinkable. Immediately apprehended, the sensible is only a semblance; but this semblance presupposes an in-itself-being. This being, an independent reality resting on itself, can be known only by thought. The otherwise material atom is here transformed into an atom of thought. Of such peculiar, positive, unchangeable thought-atoms, qualia, there is a great multiplicity. “Just as well as a mathematical point can” — in spite of *principium indiscernibilium*! — “coincide with another at the same place, just as well can indeed these absolutely positive thought-atoms penetrate one another, and coincide in one and the same point;” but such thought-atoms are

then round squares: the sensible does not let itself be dissolved.

b) The individual and the universal.

In “*Phänomenologie des Geistes*,” “this Here” and “this Now,” which involuntarily designated the punctual, were transformed by reflection into “the universal,” and the punctual suspended. But, in that by this “Now!” this “Here!” this “There!” one points to a sensible, the universal is again suspended, and the punctual comes into force. “This here!” is of course not the same as “this there!”; the universal is in the word, but the word points, by this pointing, out beyond itself to the individual. It is with respect to this of significance that Locke and Hume have investigated the origin of concepts.

In the universal concept the representation vanishes; with the determination of the individual it returns again: to mistake the representation is to disdain the fact, and to mistake the individual.

In the working-up of the element of representation two factors are active: the productive imagination and the reflecting thought. The involuntary representing forms the image in agreement with the object; hereby is conditioned the factual agreement between the circle of representation and the given surrounding world. The free imagination, on the contrary, brings about disagreement between reality and consciousness, in that it in various ways transforms the representations. If we now ask whence the thought, restlessly working on the dialectic of its concepts, gets its most pliable, most spiritual, and thus most convenient material: from the factual individual-determinations, or from the representations transformed by imagination and the freely hovering universal-images? the answer cannot be doubtful.

To what degree the objective Idealism has mistaken the element of representation by exchanging facticity’s images for hovering intuitions can be seen from many examples. “The first qualified matter is matter as pure identity with itself, as unity of reflection into itself; to that extent it is only the first, itself still abstract manifestation. Existing in nature, it is the relation to itself as independent against the other determinations of the totality. This existing, universal *Self* of matter is *Light*, as individuality the *Star*, and this as moment in a totality, the *Sun*.” (Hegel *Naturphil.* S. 130).

So long as this intuition of the nature of light keeps itself in pure generality far from all finite facticity-determinations, one may readily grant it an ingenious application of logical categories; but if, on the contrary, a connection to definite phenomena is demanded, the whole apprehension will show itself so hollow and hovering that here at

no point is there even a possibility of a penetration into the factual. Where Speculation enters more closely into a characterization of given objects, and thus engages with the individual, the misunderstanding becomes still more palpable. “Clay-earth is, like silica, the immediate, simple, unopened concept, thus the first different earthy — the possibility of combustibility... , talc or bitter-earth is the subject of salt; hence comes the bitterness of the sea. It is an intermediate taste that has become fire-principle...” (Naturphilosophie S. 416).

Instead of developing the concepts by sharp analysis of the individual, and reducing the particular in such a way that the content-rich universal can find its determinations confirmed in the given, Idealism commits the blunder of letting the universal-forms move at such a distance from the facts that these, like phenomena in the distant horizon, take on hovering outlines, and, according as “the concept” finds it serviceable, become either rocks, or animals, or clouds.

c) The real and the essential.

That one can be right in something essential and yet in reality wrong is just as comprehensible as that something can be true in general, and yet in its application to the single become untrue. That thinking is identical with being is essentially true; but that our conscious thinking and the finite existence of things should be identical is really untrue; that what exists must express the category of existence is essentially true; but that it is only the category of existence that exists is really untrue. Those who speak of essence, and those who speak of the real, speak a different language.

“The universe is to be thought under the image of a line, at whose midpoint falls $A = A$, at whose ends falls on the one side $\overset{+}{A} = B$, i.e. an overreaching of the subjective, on the other side $A = \overset{+}{B}$, i.e. an overreaching of the objective, yet in such a way that relative identity takes place even in these extremes. The one side is the Real or Nature, the other the Ideal. The real side develops itself according to three potences, in that potency designates a definite qualitative difference of subjectivity and objectivity: the first potency, A , is matter and the force of gravity — the greatest preponderance of objectivity; the second potency, A^2 , is light, an inner — as gravity an outer — intuiting of nature. Light is the higher expression of the subjective. It is the absolute identity itself. The third potency is the common product of light and the force of gravity, the organism, A^3 . The organism is just as original as matter.”

If now a strict empiricist, a man who knows of no speculation, suddenly came

to hear this Schellingian explanation of nature, he would undoubtedly regard the whole as a figment of the brain. Hereby, however, it is not proved that what is stated in those speculative propositions is in itself meaningless; hereby it is only given that the empiricist does not understand the language. If someone who would explain the world's maintenance suddenly seized Lagrange's formula: $\int \frac{da}{dt} dt = \Sigma \frac{p}{in - i'n'} \sin(in - i'n' + q)$, with the assurance that the solar system is maintained solely on the ground that i and i' can never simultaneously be equal to 0, so that the denominator $in - i'n'$ can never become 0, because n and n' are incommensurable; then even those who otherwise think rationally about real things, but have no concept of mathematics, would judge this mathematical explanation just as severely as the empiricist that idealistic one.

That ignorance and lack of philosophical sense unite to tear down Idealism means nothing; in order to understand the philosophical language, one must just as surely study philosophy as one must, in order to understand the mathematician, study mathematics. But even when we seek to understand philosophy philosophically, it cannot be denied that the objective Idealism has, with its profound cognition of essence, sinned against reality in a twofold way, partly by abstracting from real points of departure, partly by wishing to force a real content into unreal concepts. "The Newtonian theory, according to which light is to spread in lines, or the wave-theory, according to which it is to spread wave-like, as the Eulerian ether or as the trembling of sound, are," says Hegel (Naturphil. S. 141), "material representations, *die für die Erkenntniß des Lichts nichts nutzen* [that are of no use for the knowledge of light]." Hereby, then, there is abstraction from all real points of departure, and every controlling investigation made impossible.

Instead of entering into the course of the real investigations, and by the concept's dialectical superiority leading the empiricist, by force of his own acknowledged pre-suppositions, to a cognition of the essence of the matter, this Idealism has not only, with an arbitrary apprehension of such phenomena as refraction and colors, but from first to last, wished to correct the expressions and procedure of Empiricism, without, however, rightly grasping, let alone seeing through, the problems of Empiricism.

§ 10.

When it is a matter of a cognition of the real, the theory of experience must

have the decisive voice. Humean Scepticism makes, in its zeal for experience, true experience impossible. By proving the — though only subjective — universal validity of the a priori forms, Kant now doubtless brings philosophy into closer understanding with Empiricism. But, although the theory of experience to a certain degree readily grants Kant the impossibility of knowing the interior of things, it is far from grasping time and space as merely being in our consciousness. When the astronomer speaks of the moon's distance from the earth, he surely does intuit this distance from the earth, not as a merely subjective mode of apprehension, but as a real, objective spatial relation; when it is said that Neptune perturbs Uranus, the reciprocity between the planets is surely not sought in our understanding alone, but in the nature of things. The separation Kant makes between the phenomena of experience and the things themselves is a schoolroom theory that has not the least significance in the theory of experience. By asserting the objective validity of the a priori forms, Hegel, on the contrary, has — despite all his offenses against reality — in this respect pointed out the essential agreement of philosophy and Empiricism.

If this agreement is now also to be carried through in the form of the concept from the standpoint of reality, the rational and yet empirical connecting links between the a priori forms and the immediate facts must let themselves be pointed out. It is not enough that a universally valid relation between ground and consequence, conditions and the conditioned, cause and effect, and so forth, is acknowledged a priori; the task is to assure oneself of the real relations.

The relation between cause and effect is a relation of necessity that has its abiding, universal expression in law. The cause for itself is not a law, though it is an a priori law that everything must have a cause; the effect for itself is naturally not the law either; but the law is that the effect must set in when the cause takes place. On these a priori presuppositions rest observation and experiment.

When a real cause, a real ground-relation, a real law, is to be found out, one must begin with an assumption. That the earth's attraction is the cause of the stone's fall, that a liquid's evaporation is the cause of cold, that every body has its definite melting point, that ebb and flow are owed to the influence of sun and moon, all such assumptions point originally back to a subjective estimate, to impressions of experience, which, however often they may be repeated, cannot possibly prove the necessary, or establish the universal. In order then to see how far the theory of

experience is, by its “observations” and “experiments,” able to secure the universally valid, we consider especially: abstraction, induction, and the real law.

a) Abstraction.

The abstracted reminds of that from which it is abstracted; thus pure space reminds of matter, and demands, so to speak, its lost content back. To space’s discretion correspond matter’s atoms, to space’s continuity matter’s mass. The universal-determinations that lie in geometry’s lines, surfaces, and bodies have validity too insofar as they are stated of the material. That the straight line is the shortest path between two points, that two straight lines can cut each other only in one point, and so on, are propositions that can be abstracted from the material, and applied to the material. Insofar as matter is grasped as the abstract, the moment of experience here coincides with the *a priori*.

But even where the moment of experience steps forth more strongly — as when the space-filling matter is ascribed a force of resistance, a force of expansion, gravity, and so on — we are by no means, as Hume supposes, merely referred to the association of ideas and habit. “All bodies have gravity, all bodies have force of cohesion, all bodies can be divided”: a simple denial of these propositions’ universal validity has nothing to signify; they are supported by millions of impressions of experience. These propositions are indeed in their generality abstract, but this abstract is a reflex of the concrete; he who therefore will deny the universal validity must expressly adduce facts, and take upon himself the *onus probandi* [burden of proof].

What the incessant reciprocal determination of the element of experience and the *a priori* signifies is shown by physics. “It is important that the application of the most simple mathematical truths in natural science be carefully justified, so that no doubt is left about nature’s agreement with thought” (H. C. Ørsted); and it is likewise important that the propositions of natural science be grasped mathematically; for that which especially furthers the strict investigations, in which the consequence becomes recognizable on the presupposition, and this in turn on that, is precisely the exact form in which mathematics (Cf. *Philosophie og Matematik* pp. 12–14) grasps the functions in question.

The difference between the abstract, philosophical and the mathematical-physical atomic theory then also consists essentially in this, that the latter at once grasps reality in its punctuality, whereas the former never gets beyond the one-sidedly universal.

Mathematics too doubtless begins with abstractions, but it shows at the same time how the abstractions can be integrated. The mathematician speaks not merely of atoms and the atoms' union in molecules, but formulates besides the expression in such a way that it precisely suits an investigation of the real. Let the number of molecules be, e.g., m , i.e. the mass, v the volume of the mass, which encloses a number of molecules; one then has in the formula $\frac{m}{v} = \rho$ an exact expression for density, and herein a presupposition for the determination of the specific gravities of the various bodies.

If now v encloses a number of n molecules, one gets $\frac{v}{n} = e^3$, and e^3 thus as the calculating expression for the molecule. If further, in a body, m , the finite part of the mass, becomes infinitely small, thus: dm , then one obtains $\frac{dm}{dv} = \rho$, whereby then also the finite sum of the parts, Σ , passes over to $\int$, the infinite sum; one thus has for the *homogeneous* body, in that ρ is a function of the various places in the mass, the formula: $\rho \int dv$, and the formula: $\int \rho dv$ for the *heterogeneous*, and herein a fundamental concept, from which countless specifications agreeing with experience let themselves be derived.

The mathematical punctuality makes possible an analysis of reality that in sharpness can measure itself with Dialectic's analysis of essence.

b) Induction.

In abstraction the individual case is disregarded; in induction the individual gets its significance again. How the individual members lead over to the universal, insofar as this has existence in them; how the parts by their peculiarity determine the whole, so that the essence of the whole finds its complete expression in the reality of the parts – this is shown by induction. “It is the remarkable thing about this kind of inference that the conclusion is never a singular or particular, but always a *universal* judgment. The categorical inference infers from a rule to a case; the disjunctive, on the contrary, from the cases to the rule. There the rule is what is presupposed, here what is attained by inference. Categorically one can only partially, hypothetically only negatively (*modo tollente* [by the mode of denying]), infer from the particular to the universal. The disjunctive mode of inference is the only one that permits an affirmative and at the same time universal inference from the particular to the universal. Therein lies the magic power of induction: it lets us, from the juxtaposition of observations and facts, know the law.” (*Die Theorie der Induction* von E. F. Apelt. Leipzig 1854. S. 18).

With respect to the development and application of induction, Socrates, Aristotle,

and Kepler can be said to designate the turning-points.

“Two things,” says Aristotle, “can no one with fairness dispute to Socrates, induction and the determination of the universal concepts.” How Socrates proceeded, in that by consideration of a multitude of individual cases and examples he sought to clarify the ethical concepts, is well known. But, although this Socratic ἐπαγωγή is doubtless a progression from the individual and particular to the universal, it is nonetheless — as Apelt too remarks — not what we now understand by induction, but rather an analyzing art of abstraction.

As logician and natural philosopher, Aristotle is called upon to give induction a more definite form, and then sets ἐπαγωγή partly against συλλογισμός, partly against ἐπιστήμη. In the first case ἐπαγωγή is the inference from the particular to the universal, from the parts to the whole, from the cases to the rule; in the second case ἐπαγωγή also designates the beginning of the universal (ἀρχή ἐστὶ καὶ τοῦ καθόλου); whereas ἐπιστήμη only draws out rational inferences from the universal. Ἐπαγωγή leads to the principle, ἐπιστήμη develops its content from the principle. To this Apelt now remarks: “Induction (ἐπαγωγή in opposition to συλλογισμός) never leads to the principle, one presupposes it already; it leads rather to a law that ties a definite circle of phenomena to a principle. The regressive, epagogic train of thought that leads to the principle (ἐπαγωγή in opposition to ἐπιστήμη) is no inferential connection, but an abstraction. This confusion of induction and abstraction has in the history of philosophy remained standing until recent times.”

As the one who in recent times has rightly grasped and scientifically applied induction, Kepler is named; for Bacon of Verulam only emphasizes its significance, but does not know its application. Kepler’s discovery of the Mars orbit happened by application of induction; the solution of this task Apelt treats with great acumen and expertise. “The art consists in finding out, from a series of numerical values for two connected variable magnitudes, the law of their mutual dependence on one another, i.e. the form of the function. Herein lies the great difficulty of the task.” The logical form of the Keplerian induction is this: “The Mars orbit contains in itself the Mars positions:

$$O_1, O_2, O_3, \dots O_n.$$

At the position O_1 , i.e. at the time when Mars’s true anomaly is v_1 , the magnitude of

the radius vector is r_1

$$\begin{array}{ccccccc} \text{„} & \text{„} & \text{„} & v_2 & \text{„} & \text{„} & \text{„} & r_2 \\ \text{„} & \text{„} & \text{„} & v_3 & \text{„} & \text{„} & \text{„} & r_3 \\ & & & \vdots & & & & \\ \text{„} & \text{„} & \text{„} & v_n & \text{„} & \text{„} & \text{„} & r_n \end{array}$$

The Mars orbit is a curve whose polar equation is: $r = \frac{a(1 - e^2)}{1 - e \cos v}$. But $r = \frac{a(1 - e^2)}{1 - e \cos v}$ is the ellipse's polar equation. The Mars orbit is thus an ellipse, in one focus of which the sun stands."

c) The real law.

How induction, beginning with the observation of the individual, leads from the individual to the total, in that it leads through the facts to the universal, and thereby to the cognition of the real law, can be seen from the adduced example.

That the Mars orbit really is an ellipse is not proved by observing all the individual points in the orbit; such an observation is impossible, since the orbit contains infinitely many points. Nonetheless induction gives sufficient certainty. This certainty is not superficially supported on a probability-calculation that only wishes to approach the elimination of chance: no, here is an essential inference that grounds a doctrine.

To grasp the probative force of induction, one must well note that it is not to decide whether the phenomenon in question, here the planet's motion, is subject to a law or not (for that is decided a priori); but rather to establish *which* law has here in reality found its expression.

Between the a priori presupposition that here is a law, and the empirical question: which law? the distance is so great that Apelt with reason recommends "heuristic maxims" to lead the observer onto the right track. The "maxims" themselves are in this connection without special interest; whereas the following instructive direction (Anf. Skr. S. 50) ought not to be passed over.

"In the categorical inference the three termini are each time given; in induction, on the contrary, the major concept must often first be sought. When one is to seek something, one must first know *what*, and *how*, and *where* one is to seek it. Induction seeks the law on which a definite circle of phenomena depends. One must then evidently know beforehand that the law's necessity, and not chance (lawlessness), rules in this circle. Thus, e.g., the course of the stars is bound to a law, the number of

the earthly elements, on the contrary, not. But the mere presupposition of lawfulness is not yet sufficient; one must also have a certain representation of the character of the law one will seek. When Kepler sought the figure of the Mars orbit, he knew already prior to his induction, i.e. a priori, that the planets do not go their way through the heavens according to whim and chance, but that they describe around the sun an orbit that lets itself be designated according to the laws of geometry. One must, if one will not miss the way to truth, know whether the postulated lawfulness is merely geometrical, or mechanical, or teleological. Thus some seek even today teleological grounds of explanation for the phenomena of life in physiology, others physical (mechanical and chemical) ones, because one is not yet agreed about the laws prevailing in this domain.” That Galileo, who emphasizes the principle of matter’s inertia, represents abstraction, as Kepler induction, and that both directions of mind are united in Newton (*Philosophiæ naturalis principia mathematica*), is an ingenious remark of Apelt; whereas the speculative interpretation Hegel (Naturphil. S. 94–124) gives of the fall-motion and the Keplerian laws is just as unstable as his judgment of Newton’s theory of gravitation is unmotivated.

Neither the sensualist nor the idealist is able to know the real law; the former not, because he denies the a priori; the latter not, because he overlooks the fact.

§ 11.

That a cognition of the real laws is an inviolable condition for knowing the existing things, scarcely anyone will draw into doubt. But the question how a cognition of the real relates to a cognition of the essence indicates in this connection a new point of view, from which the opposition between Kant and Hegel can be more closely assessed. According to Kant it would then be possible to carry it even very far in cognition of the real, without thereby in the least coming closer to the essence of things; according to Hegel it would conversely be possible to know the eternal essence, without troubling much about the difference of the real things. How this is to be understood will first become clear to us as we more closely investigate the relation between things and laws, things and their properties, essential knowledge and real objects.

a) Things and laws.

Even when one grants what is universally valid in the laws, it nevertheless makes an essential difference whether one grasps these laws as essenceless rules without

exceptions, or sees in their fulfillment an expression of the nature of things. In the theory of experience a double language is spoken. On the one side it is said: "We do not know what matter is in itself; force is to us an unknown, we know only its expressions; the essence of light is unknown to us, we know only the phenomenon; we do not know the cause of heat, we know only its effects." On the other side it is said again: "it is not by speculations and theories, but by observations and experiments, that we penetrate into the essence of things, and get to know the hidden forces. We know nature's laws; these laws constitute the universal in the manifold, the eternal in the changes; these natural laws are true thoughts of nature, so that what we know in nature is nothing foreign, but the same as that which lives in ourselves."

These two views stand in open conflict with each other. If it is true that we know only the phenomenon, not the essence, only the expression, not the force, only the effect, not the cause; then it is untrue that the laws of things express the essence of things. If it is true that the natural laws, as revelations of an eternally active reason, express the essence of things, then the impossibility of knowing the essence, the force, the cause, is only a fiction.

Such questions as: what is space in and for itself? what is matter in and for itself? what is force in and for itself? can betray much unclarity in the a priori. Matter in and for itself, pure matter, is just as surely an abstraction as pure space. One cannot ask *whence* or *how* matter has come into space, for the being of matter and of space is in principle the same. Here is an essential identity in a real distinction; as the sensible content, matter forms a real opposition to the outer form. It is so far from the case that the intuition of an empty space should be merely subjective, that it on the contrary shows space's objective validity as a real distance between two bodies removed from each other. As little as matter can be in itself without being in space, just as little can force be in itself without being in matter. Name whatever force one will, it will always be determined by a peculiar relation in which the one matter can find itself to the other. What is force of cohesion? An expression for the resistance a body makes against the separation of its parts. Here, then, there is in the body not first a multitude of parts in and for themselves, and besides a force in and for itself; no, the force of cohesion is an expression of essence of the material parts; the strong cohesion expresses that the parts are something in themselves only in this, that they are something for one another mutually. As it goes with the force of cohesion, so it goes also with the force

of gravity, with the magnetic, electrical, chemical forces, and so on.

If all force is thought away, matter is only an empty abstraction; if matter is thought away, force is an abstraction. If one cannot say that it is matter that originally gives force existence, since matter, in order itself to be, must presuppose force; so one cannot, on the other side, assume either that force gives matter existence, since every expression of force must presuppose an existing matter from which it can proceed. One need not brood over the question how force originally came into matter, since the being of force and of matter is in principle the same.

As for chemistry's so-called elements, it is just as contradictory to wish to determine their nature in and for itself as to wish to speculate out what the chemical attraction is solely in and for itself. Oxygen, hydrogen, sulphur, such elements are only in and for themselves what they express in their mutual relation and combinations. It is not a defect in our cognition that we cannot see what the chemical force is unconditionally in and for itself; for it is, as an unconditioned in and for itself, a nothing, an abstraction, a figment. That Kant in his critique of reason had to strike upon an inexplicable "thing in itself" is comprehensible; but to wish, after all that the Hegelian Dialectic has clarified, still to hold fast to such a "thing in itself," and thereby to suppose one avoids the dialectical contradictions, is only to deceive oneself: a thing in itself, a thing behind all conditions, is a lawless thing; for things are under the law only insofar as they are with inner necessity for one another mutually, and condition one another reciprocally; a thing in itself, an unconditioned thing, is not merely a contradiction, but, when it is held fast, a hardening of abstraction, an absurdity.

If it is now evident that things lay bare their essence only in their mutual reciprocity, and if the law is the formal expression of necessity that for this reciprocity indicates the definite consequences of definite presuppositions; then it is doubtless also incontrovertible that the law is more than a universal rule without essence, that a real cognition of law is at the same time a cognition of essence, that the different laws indicate different sides of the essence of things.

b) Things and their properties.

As the essence is known by its phenomenon, so the thing must be known by its properties. The thing is the existing unity of the properties, and the properties are in turn the inner determinations of things in existing plurality. But, although the property is what is proper to the thing, the property of the one thing can in reality

become recognizable only by its finite relation to the property of the other thing; for the illumination of the matter it is required that the reciprocity of the properties be seen from the real side.

The difference that in reality is between the immediate and that mediated through many intermediate links, between the partial, which is apprehended singly, and the total, whose inner connection demands all-sided, accurate, persevering research, brings it about that the distance between a preparatory cognition of the phenomena and a concluding cognition of essence steps clearly forth.

By merely holding to the optical phenomena, one still gets no representation of the chemical effects of light; from the mere sensation of heat one cannot explain to oneself that heat expands bodies. How does it now come about that light, the same cause, can have so many and highly different effects? Can the same thing that is luminous also be warming, or is there perhaps a peculiar heat-substance? Can light be transformed into heat, and heat into light, and wherein then consists the transformation? When such questions are set in motion, it is seen that he who really wills a true cognition of the essence of light and heat can scarcely, like Hegel, overlook “the Newtonian theory” and “the Eulerian ether.”

What significance the hypotheses disdained by Hegel must have even for the cognition of essence is seen already from this, that it is a matter of an investigation of matter’s proper discretion (*Fechner: Atomenlehre. Leipzig 1855. S. 18–32*) when it is a matter of an investigation of heat’s identity with light.

That the emanation-doctrine presupposes atoms of finite size goes without saying; but neither can the undulation-theory be carried through except under a similar presupposition. In refraction the broken white ray spreads itself into a narrow fan of colors; this phenomenon can be brought into mathematical agreement with the vibration-hypothesis only when the ether-parts are assumed to be separated by finite discretion. The narrow fan of colors rests, namely, on the fact that the different rays have different refrangibility, and that the distance of the ether-parts is not vanishing in comparison with the breadth of a light-wave. The same view — that, namely, the ether-parts are in finite separation — is further confirmed when the usual light-phenomenon is compared with the polarization-phenomenon. In a usual light-ray the vibrations have all possible directions, in a polarized light-ray they are parallel, in both they are transversal. But the continuation of transversal vibrations is possible only under the

presupposition of the ether's discretion.

As now the light-phenomenon can in all respects be interpreted according to the vibration-hypothesis, so can the heat-activity too. The propagation of heat by contact is indeed so different from radiation that at first glance it should seem impossible to explain such diverse phenomena from the same general presupposition; but Fourier has shown that the phenomena let themselves be reconciled when one assumes, namely, that the bodies are not continua, but consist of separated small parts. The conductivity of the bodies then becomes dependent on the vibration-speed of the conducted heat, thus on the temperature: heat and the heated characterize each other mutually.

That the heat-rays act most strongly when they fall perpendicularly on the surface of the body, and on the contrary more weakly when they fall in an oblique direction, is inexplicable so long as one continues to start from matter's continuity; if, on the contrary, one presupposes a finite discretion, the phenomenon is not merely explicable, but even calculable, in that the ray-effect is weakened according to the law of the sine.

In chemistry and crystallography not only does the atomic theory here pointed out find peculiar confirmation, but the doctrines of light and heat at the same time undergo a new testing. If we then look at the matter in its wholeness, the essence of light can be known only from the totality of the phenomena; but the totality has an infinite compass: the one totality reaches into the other. To investigate light as the pure light in and for itself is impossible; the properties of light are known only by the properties of the things that are affected by light; the same holds of heat, of magnetism, of electricity, and so forth. These "potences" are not merely what they are, each for itself in the reciprocity of their properties with the properties of the things in question; but they are at the same time for themselves what they are in their reciprocity with one another mutually.

c) Essential knowledge and real objects.

In relation to the universal essence Hegel is right against Kant; but reality is right against Hegel. Instead of knowing the real relation between thing and properties, Hegel has already in "*Phänomenologie des Geistes*" laid out the development of concepts in such a way that in reflection one proceeds from the thing's lower to objectivity's higher category. But by this logical progression it is so far from the case that the problem of reality becomes recognizable, that it is rather veiled. The universal is the objective, but the universal is identical with thought; the objective is, in its peculiarly

determined objectivity, an object, but objectivity is identical with its concept: the object must be objectified by the subject. Along this path one never gets beyond the universal identity of thinking and being.

If, on the contrary, we let the existing, the thing with its properties, set itself in sensibly determined opposition to the subjective reflection, the problem of reality comes into force. In this point Locke sees more clearly than Hegel.

Strictly speaking, the “substratum’s” identity with itself in its many properties and accidents cannot be asserted for any real object with a priori necessity. What is this? “Silver! one sees it at once by the peculiar luster.” — But the luster can of course dazzle; another body could possibly have a similar whiteness. By one or two properties an object cannot be known; we must then, in the case of silver, test a multitude of properties: malleability, ductility, specific gravity, crystal form, melting point, attraction to chlorine, to sulphur, to oxygen, and so on. Each property must be tested sufficiently, and a sufficient number of properties must be tested: here is a task of reality in its infinity. It can be granted to the Hegelian analysis of concepts that it, against Hume as well as against Kant and Herbart, clarifies the dialectical identity of the thing and its properties, of substance and its accidents, of the concept and the object in general, and thereby a priori secures for the theory of experience an objective foundation: that the silver-substance is in essential identity with the properties of silver cannot be doubted without the law of reality being violated, and Empiricism contradicting itself; but *which* properties are here united, and *how* they are united in this real substance, must be tested by observation and experiment; the real concept fetches every moment from Empiricism.

By force of the empirical research’s inner original connection with the a priori, there is then an essential knowledge; but according to the heterogeneity of thinking and sensation, real cognition is subject to finite conditions, and all our knowledge bounded. The bound of knowledge is twofold: a *relative* one, insofar as here there is always something we do not yet know, but which we can nonetheless in time get to know; an *absolute* one, insofar as a finite separation of the content of consciousness and of reality is grounded in our organization: here, then, is something we in this existence never get to know.

II. Theory of Science.

§ 12.

Where the theory of knowledge ends in thought-confusing or merely negative results (Sophistry, Scepticism), there can naturally be no talk of true science. As the theory of knowledge is, so must the theory of science become. If the theory of knowledge draws a critical bound between phenomenon and essence (*Kritik der reinen Vernunft*), the theory of science will on all its domains remind of the bound, and as in a *reservatio mentalis* [mental reservation] always tacitly understand the impossibility of knowing “the thing in itself.” If the theory of knowledge culminates in subjective Idealism (*Wissenschaftslehre*), the scientific doctrinal structures, insofar as they are the work of the principle, will transform themselves into mirrors of reflection for “the pure I.” If the theory of knowledge raises itself to the absolute knowledge of objective Idealism (*Phänomenologie des Geistes*), the sense-things will fall, but the concepts of essence stand as dialectical realities.

On true insight into the conditions of cognition rests the true insight into the essence of the sciences. Since now the conditions of cognition are not merely inwardly subjective, but at the same time outwardly objective, the essence of science, in its different breakings-through and peculiar appearances, is besides at the same time so closely grown together with the conditions of given ages, with the aim of life, with the events of history, that in order to understand the theory one must look at the age.

What philosophy was for the aristocratic antiquity, what theology was for the hierarchical Middle Ages, that is physics for the bourgeois present: cultural history and the history of the sciences illuminate each other.

How movements in the outer relate to movements in the inner is shown by the discoveries in the 15th, and the Reformation in the 16th century. To assess the intellectual advances one must, however, distinguish between the impulses and awakenings by which great leading thoughts are called forth, and the quiet, persevering activity of spirit that tests the points of view and gives the content scientific elaboration.

Though science may at times slumber, because minds are dulled in the handed-down, and the age lacks inspiring elements; so on the other side thought can also be overstrained and the forces go astray, when the ferment is too strong and the impressions too excessive (Cardano, Telesio, Campanella, Bruno).

As for the scientific interest itself, the enthusiasm can either exclusively hold fast to theoretical ideals (Copernicus, Le Verrier), or keep closer to practical results (J. F. Böttger, Daguerre); both directions proceed from the spirit of science, for the theoretical will prove its reality in the practical.

But although science lives a vigorous life only when it finds itself in sound and true understanding with the powers of life, it cannot be drawn down into the series of means, and subordinated to the self-interested regards of finitude. Only that science is cultivated which is cultivated for its own sake. To will the scientific merely for the sake of utility is precisely not to will the scientific. Self-interest deceives itself; it wills only the means; but the means fails to appear where spirit is lacking: the mere craftsman's eye does not see so deeply that it sees the new possibilities.

Where, on the contrary, scientific insight is in every direction recognized as a spiritual power, and, without regard to self-interest's petty Wherefore?, is embraced with the interest of infinity, of ideality, science will not remain standing as a pale theoretical shadow, but, like a luxuriant tree, bear good fruits beneficial to life. The discoveries of natural science in the present century (Galvani and Volta; Ørsted and Humboldt) furnish a series of grand examples of true reciprocity of theoretical and practical interests.

A.

The Science of the Sciences.

§ 13.

Since science has its root in human life, where the ideals of fantasy go before those of thought, it is natural that the different sciences come into the world at first in

the swaddling-clothes of imagination, and are burdened with unscientific admixtures. Thus astronomy was long grown together with astrology, chemistry with alchemy, philosophy with mythology.

It doubtless lies in the drive of science to lift the pupation, break its husk, and purge out the foreign constituents. But cultivation costs time, the purging exertion; the hindrances that the human spirit has here to overcome are of different kinds.

a) An *outer hindrance* proceeds from the natural strife between egoism and truth. The intellectual love of truth has at all times waged combat, not only against the prejudices of the ages and the thoughtlessness of the multitude, but above all against their egoism, whose power and influence saw itself threatened by the rising light. So long as a given condition is of such a character that the spiritual guardians of the population have an interest in holding to the untrue, they will not easily convince themselves of the true. Science is doubtless objective; but the subjective does come into the conviction: insight is conditioned by prospect. Were there still in the countries considerable positions for astrologers, would there then not also still be those who felt themselves called to study astrology — of course not for the sake of prestige, but because they had really convinced themselves of the deeper speculative in astrology? Those who will the truth must be zealous for the truth; science too has its martyrs (Vanini, Galileo, J. G. Fichte).

b) Of a nobler, though not less dangerous, nature are the *inner hindrances* that the richly gifted, overreaching subjectivity can prepare for itself. In enthusiastic striving after deeper insight it easily happens that the thinker confounds the subjective representation with the objective object, and then supposes he finds that in reality which nevertheless is found only in his own fantasy. In the greatest minds, the noblest natures — a Plato (*Timæus*), a Kepler (*Mysterium cosmographicum*), a Schelling, a Hegel — mixtures can be pointed out, whereby partly in themselves unscientific elements are worked into scientific form, partly in themselves scientific constituents are found placed in an unscientific, fantastic connection.

c) To this comes still the tension between science and view of life, which is increased by an ever growing *conflict between the real and the ideal*.

The more science in the finite sense holds fast to the real, the more easily will it undeniably avoid the fantastic, but the more difficult will it on the other side be to satisfy the ideal demands of personality. The stronger the scientific upswing is toward

the ideal, the more easily will science doubtless satisfy the demand of spirit, but the more easily will it then also expose itself to overrunning the conditions of reality, and be caught in illusions.

What then is science? By what does one everywhere and under all shapes know the scientific from the unscientific? How ought the personal view of life to relate to the objective, impersonal results of science? For the answering of these questions is required a comprehensive theory of science, a science of the sciences.

§ 14.

From the relation between the theory of knowledge and the theory of science it follows that the true science of the sciences can be none other than philosophy. With this it agrees exactly that the true comprehensive theory of science can come about only in that the fundamental interests of the diverse sciences are represented in the deliberations from which it proceeds. Under “the comprehensive discussion and debate of all against all” (Sibbern) each science then has its vote, and each its contribution. That philosophy works up, clarifies, and organizes these contributions into an intellectual whole proves that it in this respect does not decide the matter on its own, but works as the sciences’ common organ, as their own universal consciousness, the essence that is their kinship, the light in which they know and cognize one another mutually.

It is therefore not presumptuous arbitrariness when philosophy sets the different doctrinal structures before the tribunal of the theory of science, and demands that with respect to each one of them in particular an account be given both of the foundation of the structure and of the manner of building; it is not self-taken authority when, after carefully held examination, it permits itself to assign to those theories which in foundation and manner of building have more likeness to aesthetic castles in the air than to scientific doctrinal structures “a peculiar place” outside science.

Philosophy, with all this, does only its duty, in that it, bound by the laws of cognition, does not act according to whim, but as the lawful organ of the sciences, the submissive handmaid of scientificity.

If we set together the great representatives of the development of spirit — philosophy, theology, and physics — with one another mutually; then not only can the first, insofar as it is set in opposition to the two others, by no means wish to exempt itself

from scientific justification, but reasonably neither can the latter.

Were physics, dazzled by its present popularity, to cast contempt on the investigating theory of science, and, instead of heeding the light of thought that will shine through its ground and essence, only to rely on the immediately given, and invoke the incontrovertible authority of the senses, it would only thereby set itself lower than the medieval theology. Theology too can speak of the given, use the popular, and invoke an incontrovertible authority; and so long as there is in human nature a spark of a Higher, the authoritative pronouncement of faith will always be able to outweigh the authoritative pronouncement of sensuousness.

Where, then, thus the authoritative pronouncements of Supranaturalism and Naturalism — now alternately, now partially, now in open enmity, now in mystical agreement — step into the place of the theory of science and thereby give themselves scientific importance, it is doubtless natural that philosophy is either straightforwardly despised, or at least regarded as an intellectual plaything; but then scientificity too is in danger.

But does theology then perhaps want no theory of science? If one could at once hold to Anselm and yet profit from Descartes, could get *credo ut intelligam* [I believe in order that I may understand] to say *de omnibus dubitandum est* [everything is to be doubted], and conversely, *de omnibus dubitandum est* to proclaim itself as the right *credo, ut intelligam*, then theology would surely readily unite with philosophy in commending a thorough and sound theory of science; but on these conditions it is an impossibility for philosophy to unite with theology, since the matter itself is a contradiction.

Although philology, criticism, and history — which in and for themselves belong under science — constitute constituents of theology, they by no means constitute the theological in theology. Insofar as the named disciplines are treated scientifically, they are treated untheologically; insofar as, on the contrary, they are treated theologically, they are treated unscientifically. The theological in theology is known, namely, by this, that the cause from which all effects in the realm of nature as in that of spirit are partly immediately, partly mediately derived and explained, is not sought in the nature of things or the essence of reason, but in the absolute will, in the Almighty, who creates out of nothing. But even the purest, most acute, and most consistent application of such a principle (Occasionalism; Malebranche) makes science unscientific.

When in the theory of science it is a matter of a principle, the matter cannot be decided by a More or Less, as though one, by following “the middle way,” would suppose one could still get through scientifically by assuming a little of the supernatural, if only one did not make too much of it.

The question of a principle is a question of an unconditioned, all-conditioning presupposition; such a question must therefore always be answered with Yes or No. Either theology must once and for all acknowledge the supernatural causality, and has then at the same time repelled the objectively rational foundation, or else once and for all acknowledge this foundation, and has then at the same time renounced every support in the supernatural. In the first case theology will thus cease to be science, in the last case to be theology.

§ 15.

That life and science cannot relate outwardly and indifferently to each other follows immediately from both’s intimate connection with culture and humanity; that life and science are nonetheless independent rests on a principled separation of the subjective and the objective.

In science subjective moods — inclination, disinclination, fear, hope, and the like — have nothing to signify. Insofar as the question here is of the real, we will not know how this real, according to our opinion, ought preferably to be, but how it is in itself, and necessarily must be; insofar as the question here is of the possible, the question here is not how far this possible best satisfies our hopes, but how far it agrees with the conditions of existence.

Even when science works in the service of the interests of life, it works by force of the objective. It is not subjective feelings, not pious wishes, but insight into the essence of things, that leads off the lightning-flash, and sets the steam-carriage in motion.

Science is doubtless impossible without subjectivity; but the subjectivity is serving. Here one senses, here one feels, here one thinks; but the merely subjective in sensation and thought must everywhere be purged out as the contingent, the disturbing, the untrue; every presupposition is objective. Even there where the content touches subjectivity as immediately as in the aesthetic, science sees only the true, insofar as it sees in the matter the incontrovertible, that which is independent of arbitrary

predilection, the universally valid.

From the standpoint of life, on the contrary, subjectivity rules; here all the objective is serving. In infinite reciprocity with the surrounding world personality works, even in devotion and sacrifice, for subjective interests, subjective aims, subjective ideals. Everywhere where subjectivity here is in relation to another, it is through this other in a peculiar relation to itself. On selfhood, individuality, character, personality life lays infinite emphasis. That subjectivity now, among several possible tasks, also chooses science by no means suspends the opposition of life and science. One thing it is to live for a science, another to lay science at the foundation of life.

Insofar as subjectivity, in rising conflict with the objective world, develops the personal forces, we can here distinguish the following chief stages.

a) *The relative subjectivity.* At this stage the objective is set equally as means and as end. By setting it as end, subjectivity knows its bound and relates in serving fashion; by setting it as means, it satisfies the self, and asserts freedom. Out of interest of life for the self and the work, for the worker and the labor, subjectivity will then here make manifold agreements with the objective, venture into many a strife, suffer many defeats and win many victories, in that it runs through the scale of development from the useful to the exalted, from bourgeois industry to heroic exploit.

b) *The infinite subjectivity.* By breaking through the objective barrier and overstepping the bound of the finite, the consciousness of man strives to swing itself up to infinite freedom. Since reality cannot be denied, and a material surrounding world not annihilated, the self-liberation must be ideal. In vigorous one-sidedness a choice is made between two mutually opposed expedients. Partly the objective is under all shapes so reduced to a mere means that the enjoying individuality can in each moment make itself the lord of things (Hedonism; Aristippus); partly the objective, as the universally rational, is so breathed into consciousness that all finitude is transformed into something indifferent; purified and hardened in infinite resignation, the acting subjectivity will determine itself, in that it is determined by fate (Stoicism; Zeno — Epictetus).

An inner negative infinity, a dialectical unity of ethical earnestness and carried-through irony, is the deeper presupposition and unattained ideal of these one-sidednesses. (Socrates).

c) *The absolute subjectivity.* The formal self-liberation ends in the negative; sub-

jectivity seeks life, but finds death. Only that life which overcomes death is the true, the eternal life. That the individual's innate egoism is its own original guilt, a sin that deserves death; that the pure, guiltless subjectivity has voluntarily borne the sin, and saved the guilty from death; that the natural being is sheer impotence, only the will of spirit unconditioned power; that the present is only the means, the future the end: such presuppositions demand faith, but exclude knowledge.

In open conflict with all that scientific insight can in any way acknowledge as objective, faith is so far from obtaining any support from science that it precisely by collision with the objective knowledge must be thrust back into itself, and confess that its only ground is a need of subjectivity, an inner incontestable demand. At this stage all the objective is transformed into a point of transition for the subjective. Here it is not a matter of life's relation to the lifeless, but of the living's relation to the absolutely living; here the question is not of the will's relation to the will-less, but of the willing's relation to the absolutely willing.

Life is subjective, the eternal, true life absolutely subjective; the will is subjective, the eternal, unconditioned will absolutely subjective; the objective is thus, as objective, not the True: "Subjectivity is the truth" (S. Kierkegaard).

§ 16.

In that the strife between faith and knowledge is thus driven to its extreme, the intellectual reconciliation is at the same time introduced: the absolutely heterogeneous principles cannot, in the essential sense, do each other harm. The faith that hates knowledge, because it must strengthen itself by ignorance, is a false faith. Reality will be reckoned with; it does not help to close one's eyes: if the earth really goes round, then it goes round just as well with the believing as with the non-believing. In that faith declares that subjectivity is the truth, it doubtless loses all significance for science; but, in that science declares that objectivity is the truth, it at the same time renounces all influence on faith.

Faith and knowledge can then without contradiction be united in one consciousness; their mutual strife does not proceed from the principled. The strife is psychological; it is human frailty, which has neither resignation enough to reconcile itself with science's objective, nor inwardness enough to hold fast to faith.

That faith hates science is untrue; that it makes subjectivity independent of science

is, on the contrary, true. To set faith highest and yet keep science is possible; to set science highest, and yet preserve faith, is impossible. But although science itself must see that it can neither acknowledge the absolute subjectivity nor prescribe laws to it, it is nevertheless in its interest to have this subjectivity illuminated as all-sidedly as possible.

The separation of scientific and unscientific is made difficult by this, that what according to its nature must be rejected by science nonetheless becomes an object of scientific treatment, and so in a way does go along in science. Madness, e.g., is incompatible with science; nonetheless the symptoms of madness and the treatment of the mad become an important object for psychology and medical science. As a fetus of superstitious fantasy the myth is in conflict with the scientific; nonetheless there is a science of myth, a scientific mythology.

If the most diverse can be set under scientific illumination; if even the most subjective can become objective, insofar as it is objectified in reflection; then from this point of view nothing can be objected either against a science of subjectivity forming itself, a science of the subjective.

That such a science of subjectivity (psychology, aesthetics, ethics, philosophy of religion) must not, however, be confounded with what one otherwise is wont to call a science of faith lets itself be seen when we clarify to ourselves the difference between the scientific in the wider and in the narrower sense.

Scientific in the wider sense is everything that becomes an object of science's treatment: the scientific lies in the treatment; scientific in the narrower sense is, on the contrary, only that whose validity science sees, whose truth it can assess, test, and acknowledge.

By using the ambiguity that lies in the category "scientific," the doctrine of faith sometimes succeeds in giving itself the semblance of science; it is then the task of the theory of science to point out the most important sophisms belonging here.

First sophism: "If faith is not to rest on subjective notions, and be given over to the whims of individuals, it must be possible expressly to point out what one is to believe, and in whom one is to believe: faith must have its object. To investigate the objects of faith thoroughly is to investigate them scientifically. The doctrine of faith rests on such an investigation; thus the doctrine of faith is a science."

This sophism gives the matter the appearance that the objects of faith are secured

by science, because they are tested in science. The matter is the reverse. The objects of faith dissolve into vapor as soon as science is first permitted to apply its criticism. It is just as impossible for scientific criticism to acknowledge the Gospel of the Risen One as it is impossible for science to acknowledge the myth of Hercules. To acknowledge a miracle by force of scientific criticism is a contradiction, since the possibility of a miracle makes scientific criticism impossible: where the wonder speaks, the understanding must be silent; where the miracle begins, science must end.

Second sophism: "Every peculiar science must necessarily have its presuppositions. Just as logic presupposes and justifies the validity of thought, physics the validity of experience, and aesthetics the validity of the beautiful, so the doctrine of faith must for its part presuppose and justify the validity of faith and the content of faith. He who proves that the doctrine of faith is not a science because it has its peculiar presuppositions proves at the same time that there is no science at all; but here it holds: *qui nimium probat, nihil probat* [he who proves too much proves nothing]. Thus the doctrine of faith is a science."

This sophism dazzles by the ambiguity that lies in bringing along a presupposition. What makes the doctrine of faith unscientific is by no means the circumstance that it brings along presuppositions, but precisely the circumstance that the presuppositions it brings along conflict with science. It presupposes the miracle, it presupposes the absolute validity of subjectivity and of subjective conviction; with these presuppositions it has, so to speak, forced its way in upon science, and broken the scientific measure. If the consequences are to be scientific, the presuppositions must also be so; on unscientific ground no scientific structure can be raised: natural science is possible, but miracle-science is impossible.

Third sophism: "The doctrine of faith's supernatural presuppositions have, each for itself, doubtless only validity for the believer; but between these propositions mutually there is a logical and to that extent objective connection. Since the connection is objective, and the propositions of faith precisely find their true interpretation and deeper grounding in this connection, the grounding itself is objective, and the doctrine of faith thus, by force of this grounding, a science."

The dazzling in this sophism rests on this, that a logical connection without more becomes an objective connection, and this in turn an objective grounding. "All men are triangles; Caius is a man, thus Caius is a triangle. In every triangle the sum of the

angles is equal to two right ones; Caius is a triangle, thus the sum of the angles in Caius is equal to two right ones.” Between these propositions there is here undeniably a logical connection, but is there therefore here an objective scientific grounding? From the standpoint of life there can be an eternal difference between the Holy and the Profane, the Evangelical and the Mythical; but, when science once cannot assess this Holy, cannot master this Evangelical, the difference is in the eyes of science as though it were not at all.

Since the propositions of faith are each for itself unconditionally subjective, they are in science each for itself just as arbitrary as the most fanciful fantasies. If there is then a scientific grounding of doctrines of faith, then there is also a scientific grounding of doctrines of myth, a scientific grounding of nonsense.

B.

The Particular Sciences.

§ 17.

That philosophy calls itself the science of the sciences must not be misunderstood as though it hereby would give itself out as an oracle of the sciences, the only, proper, and true science. How such a misunderstanding arises, what confusions it brings with it, ought to be heeded.

Every knowledge is determined by its object, the peculiar material that cognition works up. A comprehensive apparatus of knowledge, a methodically acquired wealth of knowledge, gives learning, but learning is not yet science; only with the development of principles and the rational working-through of the material comes the scientific. But, although to a certain bound there can just as well be science without learning as learning without science, every penetrating cognition of the connection of real facts will always require a peculiar union of learning and science. All so-called positive sciences are learned sciences.

The special is always subordinate to the universal; but in order to enrich our insight into the universal, we must begin with the special. The more a chief science expands,

the more subsidiary subjects it will secrete off. It goes with science as with Alexander's conquests; the greater the empire becomes, the nearer is the turning-point when the ring is changed and the division takes place. If now such a division, in the case of science, is not to signify the empire's downfall, but a progress of the constitution, the smaller divisions must be strictly subordinated to the greater, and these to the organized whole: the totality is the truth.

If now the totality of truth is grasped as an a priori-metaphysical essence, then it is understood from this how the philosophy of absolute knowledge has been able to set itself as the one knower upon the throne and look down on the learned sciences as on its subordinates. Learning is means, science is philosophy. The particular sciences must now either work as day-laborers, or see to learn philosophy, and explain Speculation to themselves.

"Speculation is All, is to behold in God, to contemplate that which is. Science itself has worth only insofar as it is speculative, that is, contemplation of God as he is. But the time will come when the sciences more and more will cease, and the immediate cognition set in. Only in the highest science does the mortal eye close, in that no longer man sees, but the eternal seeing itself is seeing in man." (Schelling).

But as much as these and similar proclamations testify to a fusion of religious and scientific enthusiasm, still such an apocalyptic reference to a turning-point, when "the eternal seeing itself" will become "seeing in man," cannot replace for science the lack of composure and methodical sobriety that is an inviolable condition for advance in knowledge.

One-sidedness has, as the theory of knowledge shows, its deeper ground in the misprision of the element of representation that characterizes the objective Idealism.

Just as light in given cases must presuppose a flame, the flame a combustion, and the combustion combustible materials; so absolute knowledge must doubtless presuppose diverse materials of representation, but only in order to let them burn up in the Dialectic from which the concept shines. If now the material of representation is in this way treated, not as an intellectual building-material, still less as the flesh and blood of thoughts, but only as fuel; then it follows of itself that one does not take it so exactly with the individual pieces of the fact. The consequence of this is naturally mistakes in the individual and arbitrariness in the apprehension of the total; but, even when these blunders are corrected, the standpoint will be no less false.

To value the element of representation, one should remember that the intellectual light, just as little as the physical, can furnish colors and visual images unless it is refracted in the different media, thrown back by the different objects, caught up and clarified in definite impressions. Even when the presentation of the categories corresponding to a positive content keeps nearest to the universal, one notices an essential difference between the emptiness of the barren abstractions bereft of factual knowledge and the play of colors, the natural freshness, with which concepts spring forth that are worked through by intuition, and testify to autopsy.

§ 18.

In his treatise “*Ueber den Begriff der Wissenschaftslehre*” (*Sämmtliche Werke*, Berlin 1845) Fichte has raised the question how far the theory of science can be assured that it on the whole has exhausted human knowledge. The answering of this question reduces to the following circle: “When the proposition *X* is the first, highest, and absolute principle of human knowledge, then human knowledge is a single system; for the last follows from the proposition *X*. Since now there is to be a single system in human knowledge, the proposition *X*, which really (according to the science set up) grounds a system, is the principle of human knowledge as such; the system grounded on it is that single system of human knowledge. Beyond this circle one cannot get.” Further it is said (S. 65) of the relation between the theory of science and the particular sciences: “One sees why only the theory of science has absolute totality, whereas all particular sciences will be infinite. The theory of science contains merely the necessary; all the other sciences bear on freedom, both with respect to our own spirit and to the nature independent of us.”

From the standpoint of absolute knowledge Hegel has given a presentation of the different sciences’ mutual connection.

“In a usual encyclopædia the sciences are taken empirically, as they are found. In it they are to be completely enumerated, and ordered in that the like, that which coincides under common determinations, is set together. The philosophical encyclopædia is, on the contrary, the science of the necessary connection determined by the concept, and of the philosophical coming-to-be of the sciences’ fundamental concepts and principles.”

“The sciences are, according to their mode of cognition, either empirical or purely

rational. Absolutely regarded, both are to have the same content. It is the aim of scientific striving to raise that which is known empirically to the always True, to the concept, to make it rational, and thereby to incorporate it into the rational science.”

“The rational expansion of the sciences is at the same time an expansion of philosophy itself. The whole of science then divides into 3 chief parts: 1) *Logic*. 2) *Natural science*. 3) *Science of spirit*. The sciences of nature and of spirit can then be regarded as the system of the *real* or particular sciences, in distinction from the pure science or logic, since they are the pure science’s system in the shape of nature and spirit.”

As transition from logic to natural science, mathematics is set up with respect to space and time. Natural science itself falls into: a) *Mechanics* with respect to matter and motion. b) *Physics of the inorganic* with respect to light, magnetism, electricity, the physical bodies, and the chemical process. c) *Physics of the organic*, under which then again belong geology, which treats the formation of the earth; botany and plant physiology, whose object is vegetable nature; physiology, *comparative* anatomy, zoology, which treat healthy, animal nature, as medicine the sick. The science of spirit too falls into three sections: a) Spirit in its concept, *Anthropology*, *Psychology* (language – *Philology*). b) The practical spirit, which expresses itself in deed and action; under this *the doctrine of right*, *political science*, *history*. c) Spirit’s completion in art, religion, and science, to which correspond *Aesthetics*, *doctrine of religion*, *Philosophy*.

But although the particular sciences are thus set up against one another mutually and against philosophy, their ideal independence is nevertheless only a semblance. The encyclopædia “is properly a presentation of philosophy’s universal content; for what in the sciences is grounded on reason depends on philosophy, whereas what is in them of that which rests on arbitrary and external determinations, or, as it is said, is positive and statutory, lies outside it.”

From his Kantian standpoint Apelt (*die Theorie der Induction*, S. 65–70) has indicated the principled in the separation of the sciences’ diverse fundamental character.

“There are,” it is said, “two different sources from which our cognitions spring, sense and pure reason. From the former comes sense-cognition, from the latter the pure rational cognitions. The latter fall into mathematics and philosophy. There are as a consequence three different kinds of cognitions: *empirical*, *mathematical*, *philosophical*. Each of these three modes of cognition has its peculiar character: contingency and intuitability belong to the *empirical*, intuitability and necessity to the *mathematical*,

necessity without intuitability to the *philosophical*.”

“In mathematical cognitions we become conscious of the forms of composition of things, in philosophical ones of their forms of connection *in abstracto* [in the abstract]. Kant called the former: forms of the figurative synthesis, i.e. intuitable forms of combination, the latter: forms of the intellectual synthesis, i.e. only thinkable forms of combination, laws.”

“The consideration of the different system-forms’ relation to one another and to the different kinds of cognition leads to a scientific architectonic, i.e. to a doctrine of a system of all human sciences.”

“The three different modes of cognition correspond to the three different system-forms. The pure philosophical form of cognition belongs under the categorical system; the pure mathematical form under the hypothetical system, and the pure empirical content under the conjunctive system. Philosophy is a cognition of mere concepts. But with such a mode of cognition no other systematic derivation is possible than by a categorical subordination of peculiar cases under the generality of the rule. Mathematics is a cognition by concepts of pure intuition. In this intuition the least composite is the most original and the ground from which new consequences are derived by ever new compositions. The derivation of the composite from the uncomposite happens here by hypothetical connection of thought. Finally the pure empirical has only to do with the apprehension of the individual given by sense-intuition. Each fact rests on itself, and the one stands beside the other in a conjunctive system that co-orders parts into a whole.”

The critically apt in this deduction is to be acknowledged; but it shows itself here, as everywhere, that the merely critical, without true peculiarity with the dialectical, is in philosophy unfruitful, and leads to schematism. The separation of sense and pure reason, on which this whole architectonic is grounded, has in reality no significance. The separation rests on an abstraction, but the abstracted will again be integrated by that from which it is abstracted. That sense-intuition is heterogeneous with thought is incontrovertible; but these heterogeneous are for reflection dialectical moments in the same real cognition. The sensible and the thinkable, the individual and the universal are in the thing not the same and yet the same. In reality all is concrete. The cognition of reality is conditioned by concretions; but the intellectual concretions are in turn conditioned by the integration of the abstract forms.

No one who has even merely a general concept of differential and integral calculus will, when he reflects philosophically, be able to deny the dialectical relation between the thought line, the surface's negative bound, the determinate negation that cannot be integrated, and the intuited line, the rectangle's differential, the positive bound that can be integrated into a finite rectangle (Cf. *Philosophie og Mathem.* S. 65–67; 75–76).

The immanent Dialectic will dissolve the particular sciences' independence, and transform them into moments in philosophy's system; the delimiting criticism will restrict philosophy to a peculiar rubric, and supposes on the whole to help by schemata and rubrics. Both efforts are one-sided. Though a science for itself, philosophy cannot by any critical line of demarcation be separated off from the other sciences; and although the other sciences have each in itself the universally scientific, they nevertheless do not have it in such a way that "what in the sciences is grounded on reason" should therefore be able to be said to be "dependent on philosophy."

Mathematics's difference from philosophy is not determined straightforwardly by the intuition and non-intuition of quantity, but by the peculiar analysis to which mathematics subjects the quantitative. (*Phil. og Mathem.* S. 9–14).

As for the positive sciences, their independence is immediately given with the material peculiar to each in particular, without its being thereby given that philosophy should not also in its way be able to appropriate the same material. The harder the opposition is between the object and reflection, the more remote will the positive science lie for "the absolute knowledge's" philosophy. It then makes here an essential difference whether the given object of thought is aesthetic, historical, or physical.

The aesthetic object, the given work of art, is doubtless always from one side or another related to something sensible, and must to that extent be apprehended empirically. But the empirical is here so shone-through by consciousness that the spiritual rules. The objective Idealism has then also without more vindicated for itself aesthetics as part of philosophy. Meanwhile aesthetics, with its special analyses, its autopsy, its history of art, and so on, has another side, from which it, without giving up its scientific character, may well be said to occupy an independent position over against philosophy.

That history, with its facts and its apparatus of learning, stands still more independent, is evident; nonetheless the historical object, as belonging to the past, the life of spirit, and memory, still has so ideal a tinge that Idealism itself can move within these

domains as though it were only on its own. Only physics is hard, stands unconditionally against, and permits no idealistic encroachments. Against the physical object, against reality's polygon, "the absolute knowledge" strikes, and betrays its weakness.

§ 19.

History narrates, philosophy reflects: here is a bound that cannot be leveled by a More or Less. The narration can doubtless take up reflection, and conversely; but the former is nevertheless a mark of history, the latter of philosophy.

To narrate is to determine the universal by the individual in such a way that thought is dramatized in events, and the inner goes up in the outer: here is the element of representation predominant. To reflect is to determine the universal in such a way that events furnish concepts, that intuition is deepened into the inner, and representation dissolves into thought.

From this difference in the manner of presentation a difference in the treatment of the whole and its parts is a natural consequence. The narrating history moves through the particulars of the parts step by step forward to a mediate making-visible of the whole; the reflecting philosophy begins at once with insight into the total, apprehends the particular parts as essential moments in the content of the whole, and cannot give itself over to the detail. Even when philosophy interests itself most of all in the individual, this individual is always shone-through by the total. Philosophy can from different sides illuminate the total's self-mediation through the special; but it does not tolerate the special's immediacy, and cannot follow the narrating history along mediacy's long detour. Philosophy has, as the philosophy of history, another aim than learned history; but for the same reason learned history then also has another aim than philosophy.

If here now, with respect to the aim and the means, the task and its treatment, there is a principled difference between the historical and the philosophical, then history and philosophy can neither advance in converging lines, so that they approach a fusion, according as they approach a completion. Presentations in which the historical and the philosophical equally counterbalance, and then also in a way halve each other, are only intermediate forms, neutral domains, which with their temporary fruitfulness do not concern the principles.

Learned history and the philosophy of history can mediately further each other,

correct and enrich each other, and thereby come into an ever more intimate understanding; but precisely this understanding will confirm the difference. To separate true from false, history must be critical; to point out the causal connection of events, pragmatic; to find and present the spirit whose interests and plan the events serve, philosophical. In all three respects the learned historian may well need philosophical cultivation; but his activity of spirit in no respect goes under philosophy. Everywhere referred to facts and grown together with the given, the historian's thought, far from losing itself in the philosophically universal, will on the contrary be articulated by the intuitable, and completed in artistic individuation: historiography is an art.

If a historian has made some chief part the object of an all-sidedly penetrating study of sources, he has thereby come so near to a past reality that he as it were sees it face to face; will then not his fantasy be so filled with images, and his mind so crisscrossed by individual impressions, that he can scarcely, without preceding ferment, order his material and begin his presentation of what has been intuited?

Essentially seen, the tension rests on a peculiar reciprocal relation between the real and its depicted. In the depiction the real is to be rendered; with all separation of essential and inessential there can here not, as otherwise in an art, be thought of aesthetic idealization. It is a real personality, a real state, a real age, whose physiognomy is to be seen in given features, unadorned as it is, with wrinkles and freckles. And yet spirit is to be present, and imprint itself in the depiction, that it not be overloaded with the trivial, and run out into the real's endless prolixity.

With such a task in view the historian will see that there is no peculiar form that completely exhausts reality's content, but that in the presentation there is incessant strife between loss and gain. What here in one respect is gain, is in another respect loss; this holds not only of the plan on the whole, but repeats itself even in the execution of the individual. That the party spirit does not always need to do violence to the facts in order to get history on its side, it owes doubtless nearest to its own sophistical suppleness, but yet also essentially to the circumstance that the objective in history, however strictly it is held fast, necessarily obtains a coloring from the objectifying consciousness. Different presentations of the same subject can, according to the individuation, be equally justified.

But if the historical form is so artistically individual that one and the same universal task (e.g., the history of the French Revolution) first finds its complete solution when it

is solved in several peculiar ways; then it is seen from this to what degree the element of representation must protest against a fusion of learned history and the philosophy of history.

§ 20.

It is not the whim of chance, and not human arbitrariness, that may draw the bound between the particular sciences. Each science is in itself an essential totality; its bounds, divisions, and elaboration are conditioned by the material, governed by thought, and determined by the comprehensive task.

In the method of every science the universally scientific is seen: everywhere a reciprocal determination of the immediate and the mediate in cognition, of detour and shortcut, of necessary movement of the matter and free movement of thought (Philosophie og Mathematik. S. 44–46); likewise every science's systematizing striving will require definitions, divisions, subordination of lower and higher unities under one another mutually and under a highest unity, and so on. But the peculiar material and the peculiar aim in which it is worked up will nonetheless in the method as in the formation of system bring with it such decisive peculiarities that each science thereby not only becomes a doctrinal structure in a peculiar style, but is even in an inner, infinite way organized into a relative monad, an intellectual individuality.

To what degree the peculiar material acts upon the formation of system will especially become manifest from those sciences whose content is expressed in typical natural forms. Here is, in the mineral kingdom, the dead form: the crystal; in the plant kingdom the living form: the organism; in the animal kingdom the ensouled form: the individual.

a) *The dead form.* In its typical purity the substantial form of bounding is mathematical. By laying a mathematical body (e.g., the cube) at the foundation for the determination of the axes and the corresponding section, pure crystallography can with the sureness of an exact science round itself off into a whole articulated into particular systems — the spheroidal, the rhombohedral, the pyramidal, the prismatic. Hereby it becomes not only possible to lead an infinite manifold of derived special forms back to their fundamental form valid in each system; but even the combinations, alterations, and distortions that proceed from formations in free nature are so far from bringing the whole into confusion that they on the contrary become intelligible by

the systematics of the pure types (Applied crystallography).

It is otherwise when the question is not merely of the substantial forms of bounding, but at the same time of the corresponding substances' chemical and physical properties, of the mineral's characterization. Here the preponderant regard to the chemical substantiality (Berzelius) comes into open conflict with an equally preponderant regard to the individuality expressed in the physical properties (Mohs), the strict demand for systematic grounding in conflict with the schema and its many rubrics.

A more equal regard to chemical and physical distinguishing marks can now doubtless lead to ingenious combinations of system and schema; but only gradually, as science penetrates into the essential connection of the substances' chemical and physical properties, will the schematic go up into the systematic.

b) *The living form.* Although the real bound that separates the organic from the inorganic does not immediately let itself be pointed out, the essential difference between the mineral and the plant is nonetheless clear enough. Whereas the crystal, the dead form, goes up in the mathematical, the mathematical (three-numbered plant, five-numbered plant, the difference-angle of the stem-leaves, and so on) is, on the contrary, as mathematical of subordinate significance in the living forms. In the different species as well as in the individual specimens of each species the organic runs through such a series of metamorphoses and corresponding articulations that here one can rightly speak not merely of incessant metabolism, but also of lower and higher stages.

The sexual system, the so-called artificial plant-system, with its classes and orders (Linnaeus), is a grand, comprehensive, on the whole practically applicable schema; but only when a plurality of essential properties and organ-relations is taken into the grounding of the division does the natural, i.e. the proper, system begin to form itself. Strict holding-fast to a metaphysical principle and arbitrariness in the apprehension of the given (Reichenbach) has, however, instead of bringing the system to completion, on the contrary shown what difficulties are here still to be overcome.

c) *The ensouled form.* Whereas the individual unity in the case of the plant either is not at all, or is at least so protean that it vanishes into phytons and cells, the animal is, on the contrary, by force of the concept's necessary demands (sensation, will), originally set as a definite individual. The stricter articulation, whereby the organism's individual members are developed for themselves, advances on the whole

with a rising centralization, which gathers the organs' different activities into the sensing, self-determining unity of life.

It is then a matter of course that the distinguishing marks of the species and the typical separation into lower and higher orders — *molluscs, articulates, vertebrates* — must in the animal kingdom appear with a quite different clarity and orienting validity than analogous divisions in the plant kingdom. Strict Empiricism (Cuvier) has here too stepped into opposition to natural-philosophical Speculation (Geoffroy St.-Hilaire). But, in order that the system can be brought to completion, the reality of the genus-concept must first be decided, and the connection of the higher classes with the lower determined in such a way that chief forms and intermediate forms show themselves as naturally necessary members in the ascending chain of ensouled unities.

If now the universally scientific is everywhere by material and task grown together with the peculiar, then each science must also exclusively look to its ideal, and cannot possibly take its scientificity in fief from philosophy.

How little philosophy is entitled to ascribe to itself the thought and the developments of concepts that belong to the learned sciences is expressly seen from this, that categories such as substance, individual, species, genus, and the like are developed in the real sciences in a quite different way than in an objective logic or speculative natural philosophy.

The laws of thought are doubtless everywhere the same; but to hold fast to the laws of thought for themselves in their pure generality is precisely to keep them in a questionable one-sidedness. The strict thinking has in the one science a quite different character than in the other: the particular sciences presuppose particular talents.

§ 21.

Insofar as the particular sciences have in material and aim a finite delimitation, they are subject-sciences [Fagvidenskaber]. How peculiarly the different subject-sciences' view is formed becomes manifest when they, by treating one and the same subject, approach mutual contact. In the determination of soul, life, and life-principle a thoroughgoing difference shows itself between the strictly empirical physiology, for which the organism is "a living machinery" that does not even need a life-force for "machine-master" (G. Valentin), and the psychology grounded on self-observation,

which grasps the soul as constituting essence, and lets life well up from “an inner source.”

The strife that can thus arise between real subject-sciences must, however, above all not be confounded with the strife between the theory of science and a miracle-science. One thing is science’s strife with the unscientific; another is the strife between scientific one-sidednesses. One thing is a strife against confusing appendage of an unwarranted one; another a strife for true equilibrium between warranted ones. One thing it is that the archaeologist and the geognost, each equipped with his apparatus of factual proofs, meet in strife at a burial-mound; quite another that a medieval theologian, without acquaintance with the natural marks, invokes a supernatural, and mixes into the investigation about the human races and “our first parents.”

In order that the individual science can advance, it must be supported by those most nearly related to it. In that mathematics is a chief science for itself, it is at the same time an auxiliary science for mechanics and astronomy. Chemistry, mineralogy, and geognosy unite as auxiliary sciences in geology. The helping sciences are in turn helped by those they have themselves helped. To what degree botany and zoology further geology, and are in turn furthered by it, is conspicuous. In the great circle of coordinate tasks no subject has an exclusive title to be the chief subject: in the organism of the sciences the means is aim and the aim means. As easy as it is to grasp the particular sciences’ mutual reciprocal relation in general, just as difficult can it in turn be to determine this relation with respect to the individual science in particular. Physiology furnishes a telling example. The doctrine of digestion presupposes insight into the chemical processes; how many constituents ought now to be taken up from chemistry? The doctrine of sight presupposes optics, the doctrine of motion mechanics; how much ought now to be borrowed from these sciences? Thus one could go on.

Traditions and conventions have in such cases no decisive voice. Every expansion and restriction must proceed from the nature of the matter. But precisely in the development of the matter lies the difficulty. The one science can doubtless refer to the other; but in the mere reference the relation is external. If the point to which reference is made lies outside the matter, then the reference is, strictly speaking, superfluous. If, on the contrary, the point in question is an essential part of the matter itself, then it ought also to that extent to be taken up into the development of the matter.

But in what form is it to be taken up? As a borrowed proposition? But the borrowed

proposition is like a dead, mechanical admixture, a torn-off branch that is attached to a foreign trunk. If the attachment is to be organic, if a grafting is to take place, then the borrowed proposition must bring along so much of the grounding connection that thereby a part of the one science is carried over into the other. But in this way the one science will come to take up a multitude of extracts from other sciences; and yet every *juxtapositio* [juxtaposition] is in conflict with the essence of science.

The diverse contributions a science receives from others it must assimilate; references, borrowed propositions, extracts, all constituents conveyed from without must be worked up in such a way that the externality vanishes, and the patching-together is suspended. To suspend the patchwork and work together the heterogeneous contributions requires now doubtless a peculiar organizing talent, but then also above all a clear and distinct insight into the relation between subsidiary subject and chief subject. This relation becomes especially involved when sciences that in and for themselves are chief subjects partly furnish a comprehensive foundation, partly, for a definite purpose, are treated as subsidiary subjects.

Mathematics, which in itself is a chief subject, and for astronomy and mechanics a comprehensive foundation, must, insofar as it serves chemistry, be treated as a subsidiary subject. Chemistry, which in itself is a chief subject, and for mineralogy and physiology a comprehensive foundation, must in many peculiar cases expressly be treated as a subsidiary subject. The category foundation indicates in this connection an intermediate determination between chief subject and subsidiary subject.

To treat a science as a chief subject presupposes, however, only one decisive regard; but to treat a chief science as a subsidiary subject demands a plurality of regards, a doubling of reflection that makes the task difficult.

The bound between the different sciences is in its essence dialectical; but in the subject-sciences this dialectical cannot become the object of any peculiar exhaustive reflection; instead of such a reflection there steps a living leading thought, a peculiar decisive purpose, a definite plan. The aim expressed in the plan is a fundamental concept determining all peculiar demands, penetrating all parts of the presentation. Where this fundamental concept is not clear, where the definite aim is not at all points present to consciousness, the presentation will become now too prolix and overloaded with unnecessary detail, now again — of course “for lack of time!” — too superficial and survey-like.

The principles according to which the branches of science, with their chief and subsidiary disciplines, are grouped under a university's different "faculties" cannot possibly act solely according to abstract-theoretical regards; for that science is in an all-too-many-sided relation to real life.

The reciprocal relation between science and life is mediated by the state and its institutions. In the different faculties several often very diverse sciences are worked together into peculiar wholes calculated for definite spheres of activity. If these circles of knowledge are rounded off according to all-too-meager practical regards, science will be violated, and the scientific spirit lost. If these wholes are organized according to all-too-high-flown theoretical ideals, the interests of life will step into opposition to the scientific, and science lose.

The reciprocal relation between theory and practice must stand clear to consciousness as the plan is laid. Once the plan is laid, and the means assigned, the carrying-through ought to happen with strictly persevering consistency. The interest now falls decisively on the theoretical, precisely because this theoretical has taken up the regard to the practical as a moment integral to its plan. If the study of theory can really ground a station in life, then it also deserves that fresh vital forces be applied to bring it to its best possible completion within fixed bounds. The student who in this respect, without doing his duty, would filch the final advantage, is false toward science and false toward the state. For the state will not merely have a number of businessmen as practically usable organs; it will in these organs see the scientific spirit, the higher cultivation, represented.

C.

Philosophy and the Particular Sciences.

§ 22.

When the ancient problem how a science is attained is set forth thus: "The knower can, insofar as he already is knowing, get nothing to know; but neither can the ignorant

one get anything to know, since he, in order to come to a knowledge, would first have to see his ignorance, and, in order to see his ignorance, already beforehand be a knower,” the problem must doubtless at first glance look like a sophism. The matter nonetheless has, in the unconditioned sense, its correctness. One can, as regards finite expertise, get much to know that one did not know beforehand; but from an absolute not-knowing to knowing there is no finite transition.

As for the *a priori*, Plato (Phaedrus) has already pointed out the essential (ἀνάμνησις). Every advance in knowledge is point for point conditioned by an immediately evident; since now nothing can be evident except in that it goes together with our own consciousness, and reminds us of what we already know, the advancing development of concepts rests on an intellectual recollection, a reflecting recall, in which we incessantly find the True as in ourselves, and, so to speak, take our own knowledge into possession.

He who learns mathematics gets essentially nothing else to know than what he originally knows, and yet by learning mathematics he precisely makes advances in knowledge; the same holds of the *a priori* in aesthetics, ethics, and dialectic.

With respect to the *a posteriori*, the advance of knowledge will doubtless be conditioned by this, that here new knowledge is constantly gathered, that new facts are tested, new discoveries made, and so on; but is the matter herewith decided? If the advance of knowledge in relation to the *a posteriori* is grasped solely from this side, it is easy to dismiss philosophy. If it cannot even exhaust the *a priori* — since mathematics owes it no thanks — how much less can it then engage with positive facts! If the historian sets it a task: it cannot answer, for a historical task is answered only historically. If a chemist turns to it: it has no knack for experimenting, makes no discoveries in chemistry. If the astronomer wants help with an observation: it scarcely understands him. If the physician asks it for advice: it is ignorant. Even what it has on its own speculated out concerning the relation between soul and body, the physiologist perhaps cannot with difficulty even use.

But in these affairs too it is a matter not merely of getting something new to know, but also of knowing what one already knows, of getting one’s knowledge thoroughly, inwardly, all-sidedly brought to consciousness.

Although every science that begins with the first principles, and goes systematically forward, doubtless develops its content in such a way that the otherwise prepared

beginner can follow; so it is nonetheless unmistakable that one first fully understands the beginning and the earlier when one, after having come to the conclusion, looks back. The recall whereby the foregoing is illuminated by the following is an inwardizing of the matter, a doubling of reflection, which, with respect to finite points, does not indeed enrich us with anything new, but nonetheless in infinite degree heightens our knowledge.

If there is now, also in relation to facts, an advance of knowledge that consists not in a multiplication of the things that are illuminated, but in the intensity of the light that illuminates the things; then it depends in this respect likewise on reproduction, on inwardizing, on recollection (ἀνάμνησις), and thus philosophy has to that extent the same validity in the *a posteriori* as in the *a priori*.

In positive matters, in the affairs of experience, philosophy does not go before, but follows after. Everywhere where Speculation with its “intellectual intuition” has wished to anticipate Empiricism, lift the mystery, and unveil the hidden, it has only made itself ridiculous. But if anyone now concludes from this that philosophy is just as superfluous when it follows behind as it is unwarranted when it will go before; he can, by dwelling on what he already knows, without difficulty convince himself of his mistake.

To dismiss philosophy one could adduce that the higher knowledge owed to afterthought’s reflecting retrospect is attained not by philosophizing, but by repeated study of the subject-sciences in question. A deeper insight into the essence of mathematics is won not by philosophizing, but by incessant repetition of mathematical analyses; a higher consciousness of the chemical is won not by speculating, but by again and again revising what is won, and making the recognized truths quite living, penetrating, and clear to oneself. By thus going through science after science, everywhere with sharpened eye for presuppositions, principles, and the inner infinity of a systematic connection, afterthought will then bring the heightened light, and enrich us with an insight that can disdain philosophical abstractions.

If now this yield of consciousness is brought out for itself, in such a way that it can take on the shape of the concept and be clarified in categorical form; then we have full compensation for the vanished philosophy.

But in this compensation for the vanished philosophy, philosophy is precisely present. Every relation to an object is in science an objective relation of consciousness.

One can by consciousness set thought above sensation, the ideal above the real, but not deny consciousness's barrier. One can by consciousness set sensation above thought, the real above the ideal; but one cannot by consciousness set anything in consciousness's place. To become thoroughly, all-sidedly, inwardly conscious of all real and all possible relations of consciousness is philosophy's task. In this its unity of essence with the knowing consciousness, philosophy is precisely present when one proves its non-being, precisely in exaltation when one sees through its degradation, precisely vigorous when one considers its weaknesses — a spiritual power, a superior potency, an infinite Proteus invincible to the subject-sciences taken together.

That positive knowledge does not without more replace philosophy becomes not so seldom evident when learned researchers, especially natural researchers, give up the strict method, write popularly, and impart to the cultivated an improvised philosophy.

§ 23.

Over against the other sciences philosophy is doubtless a particular science, but nonetheless no proper subject-science. Although such ideal objects as thought, truth, life, the good must be said to be philosophical; thought is nevertheless not an object for philosophy in the same sense as the starry heaven is an object for astronomy, the plants for botany, the animals for zoology.

The subject-sciences are solid, always have fixed work, ample livelihood, and secure recognition in the learned republic; philosophy is in comparison with this like an itinerant without finite provision, a hovering shape that sinks and rises, dies away and lives up again, according as the spirit wills it. The subject-sciences are the rich landowners who have the objects, and live in the manifold of knowledge; philosophy is like that Eros, born of Poros and Penia (Symposium, Plato), always in want, owns nothing, brings nothing but the gift of penetrating everywhere, and being sated with a glance at the abundance of the rich.

He who cultivates a subject-science has, with all his exertion of spirit, nonetheless always an unmistakable advantage in that his material has a finite bound. It is otherwise with him who will cultivate philosophy. Instead of enjoying the impression of independence, the intellectual ἀντάρκεια, which persevering occupation with even merely a single branch can grant the capable specialist, a continued study will on the contrary convince the philosophizer that he cannot take a step on the path of

real cognition without feeling his knowledge's dependence on the learned sciences; whereas the specialist will soon learn to help himself as an independent researcher, it can be the philosophizer's humble position incessantly to knock on a foreign door, incessantly to begin anew, and so be an apprentice his whole life.

With all its idealistic mobility philosophy is nonetheless a science determined in the concept. If each of the subject-sciences can teach us how it will be studied, then philosophy too can for its part offer a similar guidance.

It is natural that the study of a subject-science is united with the study of the science's history; but insofar as here there is an immediate material, the treatment of the material will step into the foreground. One then readily begins by studying first a part of natural science itself, and next the history of natural science; first historical events to a certain extent, and then the history of historiography. Science itself will in such cases stand above its own history.

In philosophy the relation is reversed; here the history of the science is an essential content for the science itself. The history of philosophy in union with the philosophical chief works then forms the foundation with which here one must begin.

The next step is the study of the real sciences. To behold the works of art, live with the poets, and make oneself familiar with the historians requires general cultivation, but no peculiar guidance. This is, on the contrary, the case when the talk is of the proper real sciences. That insight into the nature of things is attained only by real investigation of the natural things is shown by the theory of knowledge; but the mass of knowledge that must here be presupposed is so great, the all-sidedness and the compass in which the studies in this respect would have to be driven so considerable, that one, without looking to how much a single person can accomplish, may only look at the science itself.

That a philosophizer should be perfectly at home in all possible sciences, and, like Aristotle in his day, possess a whole age's apparatus of knowledge in all its particulars, is not only impossible, but also in the matter itself superfluous. The philosophical analyses of reality are doubtless in so intimate connection with the facts of experience and corresponding observations that handbooks, surveys, popular lectures, and the like do not suffice; but it is nonetheless also a law that philosophy must not take up more material than the Dialectic can dissolve, and the concept assimilate. It then depends not so much on the greatness of the mass of knowledge as on choosing among

the mass's parts, sifting the material, and, so to speak, finding the ore from which a knowledge can be smelted out.

As an inviolable presupposition the theory of knowledge must demand that the philosophizer, even if he is not a mathematician, nevertheless learn to understand the language of mathematics, and, although he will not become a chemist, make himself familiar with the chemical laws, designations, and formulas. Without such a preparation the philosophizer will never penetrate into the kernel of the matter, never see through the exact adducing of proof, never understand what physics really means when it speaks of force and matter. Mathematics and chemistry are, so to speak, the keys to the secret recesses of real science; only he who has these keys is an initiate, and only for the initiate does the treasure show itself. In order that a study of so many and different branches of science shall now not either lose itself in bottomless thoroughness, or pass over into loose diletantism, it is naturally necessary that the philosophizer be conscious of a task, that he in all this see a definite goal, and have the endurance to pursue it.

As for the task in its generality, it will be determined by the reciprocal relation between the real and the ideal, between analysis of representation and analysis of concepts, between the principles and the finite presuppositions.

As for the task in its peculiarity, partly an individual question arising under peculiar circumstances, partly a comprehensive regard to philosophy's organic chief parts can determine the compass and direction of the real studies.

But even when the task is held fast, even most strictly, the real studies still bring considerable difficulties peculiar to the relation with philosophy. For an analysis of concepts it is namely not enough that the philosophizer knows a number of the points essentially attractive to thought; he must have such points in such a manifold, possess them with such freedom, and move with such sureness, that he, without striking against the real, can move his moments of concept as though they were not piecemeal-gathered contributions, but inventions of his own fantasy.

How difficult it is to acquire such sureness can easily be seen. From many sides Empiricism illuminates the inner, mysterious connection that is between things' otherwise so different properties. The universal is here known in the special, but this special in turn interrupts the cognition of the universal. Let one, e.g., order the elements in the so-called trilogies; let one compare barium (*A*), strontium (*B*), calcium

(C), and note the relation between the chemical numbers: $\frac{A+C}{2} = B$. If one now looks at the solubility of baryta, strontia, and lime, then one finds that the baryta has the easiest, the lime the heaviest solubility; whereas the strontia stands midway between. Let one test another trilogy: sulphur (A), selenium (B), and tellurium (C), whose chemical numbers will likewise give: $\frac{A+C}{2} = B$. If one now looks at the specific gravities, these relate thus: the sulphur has the greatest, the tellurium the least, and the selenium stands midway between. Let one think of the relation between the specific heats and the chemical numbers; let one think of the crystals' relation to light and their expansions under heating; let one think of the law according to which a rise and fall of the boiling point of the polymeric hydrocarbons (CH_2 , C_2H_4 , C_4H_8 , $\text{C}_{16}\text{H}_{32}$, ...) takes place, in that the point rises for each carbon atom that is added, and sinks for each added hydrogen atom, and so on.

But while here, under this penetration into the interior of things, a multitude of parallels, a series of presuppositions and consequences, come to light, there can nevertheless in this connection scarcely be made a single a priori inference independent of the test of experience. That the specific gravity of iron oxide is less than that of iron is comprehensible; but, if one drew from this the general inference that the higher metal oxides on the whole must have a smaller specific gravity than the lower, one would err: the specific gravity of stannic oxide is, e.g., greater than that of stannous oxide; that of plumbic oxide greater than that of plumbous oxide. If we look at the relation between the two properties, hardness and specific gravity, then it seems natural that on the whole they should correspond to each other: the diamond, which is far harder than graphite, really also has greater specific gravity; yet let one guard against a hasty inference.

When isomorphic substances vicariate, it is seen that hardness and specific gravity by no means straightforwardly correspond to each other. If iron steps into the garnet instead of lime, or lime instead of iron; then it holds: the more iron, the greater specific gravity, but conversely: the more lime, the greater hardness. Every essential connection doubtless contributes to the answering of a Wherefore; but here comes every moment a Wherefore that cannot yet be answered.

For the philosophizer the subject-knowledges cannot in and for themselves have significance, and yet it is not so seldom that precisely those points which are quite properly fruitful for the analysis of concepts lie hidden under a detail of facts and

special investigations, which thus must nonetheless be gone through.

Finally a deeper appropriation of the intellectual that offers itself in a definite circle of facts can first set in when this intellectual is illuminated from other and corresponding circles: *Opposita juxta se posita magis illucescunt* [opposites placed side by side illuminate each other more].

If the total can never be fully developed except by means of the individual, then the reverse also holds, and first and last in philosophy, that a penetrating study of the individual cannot possibly be carried through without a corresponding insight into the total.

§ 24.

But even if the philosophizer has now also conquered the difficulties of real study, drawn sublimates from the positive sciences, and sees himself able to work together the constituents: in what form is he then to impart what has been won? to whom ought he to turn, by whom will he be understood? For the illumination of the difficulty here pointed out we grasp the matter, first from an objective side, in that we look at the manner of presentation, next from a subjective side, in that we take account of understanding.

The manner of presentation. As for facts and corresponding laws, philosophy must treat the given as a given, and not by a kind of sophistical turn let the matter get the appearance that things which are due solely to experience nonetheless proceed with inner a priori necessity from the pure concept. The untruth that in this respect crept into the Hegelian method has doubtless its ground not in any conscious dishonesty, but in overstrained zeal of reflection and a consequent disproportion to the immediate.

Immediacy ought doubtless to be lifted; but immediacy must not be suspended in such a way that the given changes its nature, and is transformed from a positive fact into a metaphysical essence; immediacy must be suspended in such a way that the given, as given, furnishes an ideal contribution to the comprehensive development of concepts that the principle itself sets going. Insofar as the comprehensive development of concepts is and remains the main thing, the given then lays off its contingency, does not remain in thoughtlessness, but shows its ideal, obeys the concept, and comes when thought calls.

The point at which the conceptual side of the given each time coincides with

the leading concept of the development is a point of insight. At this point there will always gather so much light that the proposition which is taken up as a given from another science neither presents itself as a dead borrowed proposition, nor brings with it, for its final explanation, a circumstantiality that halts the pace of development and burdens thought with the all-too-mediate.

To conduct separate proof for each peculiar proposition is impossible and unnecessary; to illuminate each proposition in such a way that the side that is essential for the connection is also clarified in the connection, that the point on whose understanding it each time depends can also each time be understood, is possible and necessary.

Every scientific presentation of deeper content always has a multitude of luminous points in the foreground, a multitude of obscure points in the background, and between both manifold transitions. Where several sciences treat the same thing, each will make its own distinct; but they will never make the same distinct in the same way. Likewise with respect to clarity. What one development according to its purpose must lift to clarity, another may according to its purpose let remain in obscurity. The logical categories obscurity, clarity, and distinctness are of dialectical application.

Another difficulty with respect to the given is peculiar to philosophy. This science may not, as a positive subject-science does, give itself over to the manifold of the finite, enumerating, narrating, describing, and so on. Nonetheless it must in its way take up the outwardly manifold, insofar as it will give itself a real content.

To remedy this difficulty one has sometimes seized on the desperate expedient of laying out a schema of concepts, holding to the formal order, letting the reasoning within each of the rubrics run free, and then filling the emptiness with a multitude of heterogeneous examples fetched from all positive sciences; this is perhaps the worst method anyone can apply.

The task is to let the form take up the content in such a way that the development of concepts becomes a development of the matter. In this development of the matter the positive elements must then be chosen, distributed, ordered, and treated not merely in such a way that the individual facts are placed where they are singly needed, but also in such a way that the whole aggregate of real moments is worked through by reflection, that each concluding totality can express the content of the matter in the form of the concept. Through concluding totalities in different — partly coordinate, partly mutually enclosing — circles the development will then go to the last conclusion.

Understanding. Where tradition and habit once give a certain manner of teaching prestige and support, it is often striking to what degree one can suppose one explains the inexplicable and understands the incomprehensible (half-scientific religious books); where, on the contrary, the principle of cognition demands that in something handed down several changes necessary for understanding must be made, it is again just as striking to what degree obstinacy can strive against, and find the in-itself-understandable incomprehensible. That opinions about the understandable and the incomprehensible can be so divided, that one can make such highly different demands on what is to be understandable, follows from this, that the category understanding is so infinitely dialectical.

It is an easy matter to reproach a philosophy with incomprehensibility. General cultivation does not as a rule go deep enough to follow a stricter analysis of concepts: the understanding of the individual presupposes expertise. But if special expertise is an inviolable demand, then the consequence of this must be that each specialist holds to his piece, and lets the rest stand as an incomprehensible. That such a parceling-out of understanding must make philosophy meaningless is then understandable.

As for the presentation, it can become incomprehensible just as well by holding to the general as by going into the special. If special analyses are predominant, then one doubtless notices the involved, the thorny, the prickly; but if the universal-forms are predominant, then one gets perhaps still more troubles with “the abstract,” “the nebulous,” “the hovering,” “the ambiguous” (turns of phrase, word-philosophy, phraseology). It depends in such things not on the finite What, but on the infinite How.

To clarify to oneself the relation between understandable and incomprehensible, one must, on occasion of the demands that can with right be made on the presentation of what is to be understood, then above all also not forget the demands that can with right be made on him who will understand, such as: necessary preparation, persevering attention, scientific sense.

Insofar as each science has its peculiar structure of thought, it consequently also has its peculiar kind of understandability. In the mechanical, understanding comes otherwise than in the chemical, in the exact otherwise than in the dialectical, in mathematics otherwise than in philosophy.

§ 25.

If we call investigations of things' first presuppositions and knowledge's principles philosophical, there can doubtless be no doubt that philosophical investigations are necessary in the different sciences; but from this it does not yet follow that philosophy for itself is a peculiar science. One must, e.g., in every branch of science apply criticism, and also has in the different sciences a multitude of critical treatises, even treatises on criticism; but for that reason surely no one will maintain that criticism for itself is a peculiar science.

If there were now a circle of critical-philosophical treatises in which the diverse sciences had each in particular given an account of its principles and presuppositions; then one could doubtless, by working them all up according to one plan, unite them in a whole; but this whole would, by the finite connection of the parts, nonetheless just as little become a peculiar philosophical science as the collected treatises would become a peculiar science by being bound in one volume.

The objective Idealism would now at once lift philosophy's scientific independence by referring a) to the Dialectic in which the finite contributions are dissolved, melted down, and deposited as the mobile moments of an inner infinity; b) to the immanent method whereby the absolute content gets its moments laid out, separated, and gathered in the absolute form; finally c) to the deeper view of life, the solution of the riddle, which is attained only in Speculation.

But the presuppositions on which philosophy not long ago wished to ground its divine superiority unite here as flaws that threaten its scientific independence. Thus it holds with respect to a), that the dialectical dissolution may nonetheless not loosen principles and presuppositions from the soil of reality in which they have their root; with respect to b), that "the immanent method" must go the same way as "the absolute knowledge"; and finally with respect to c), that science does not concern itself with "the deeper view of life," since it on the whole does not sit in judgment over views of life, and that "Speculation" cannot solve "the riddle of existence."

How far philosophy is a science independent in itself is not dependent on arbitrary opinions or passing blunders, but on the nature of the matter. If philosophy is a science, then the peculiar form in which it develops its content must be of such decisive significance that scientificity would suffer an irreparable loss if there were no

work on bringing this form too to systematic completion. We must then here expressly bring out the influence philosophy exercises on the scientific life of consciousness, insofar as it is a matter of the appropriation of the material, the liberation of the concepts, and the inwardizing of the matter.

1) *Appropriation: an assimilating activity.* The outwardly given does not immediately go up in consciousness. In every subject-science there will always be a mass of constituents so heterogeneous with the life of consciousness that not even the strongest memory can gather and bear their quantity; such constituents are then preserved as materials in magazines, repertoires, and the like learned storerooms.

In relation to the real, scientific insight is conditioned by memory; in relation to the essential, by recollection; the real is remembered, the essential recollected. The appropriation under which the scientific material passes over into the flesh and blood of consciousness (*in succum et sanguinem* [into sap and blood]) is an intellectual process of digestion that works up the material, and takes on a different character according as it is the real in its particulars or the essential with its universal-determinations that becomes the main thing.

The more the side of reality predominates, the more closely thought joins itself to memory, and the more also the appropriation becomes restricted to a peculiar rubric of consciousness, determined for the peculiar subject. The more, on the contrary, the side of essence is brought out, the more intimately thought goes together with recollection; the finite constituents of the material then become real moments in the peculiar concept, and the appropriation of the peculiar concept sends out a light that shines through the whole consciousness.

When now these two modes of appropriation are referred each to its aim and its principle, a difference shows itself between subject-science and philosophy.

The carried-through assimilation relates to Speculation as real-philosophy to Idealism. In idealistic Speculation the given is volatilized into a priori essence; in real-philosophical assimilation the a priori essence is peculiarly determined by the given.

2) *Liberation: a humanizing activity.* Restricted to the different rubrics of consciousness, the won concepts are still so grown together with finite constituents, still so burdened with particular presuppositions, that they find themselves as in a state of unfreedom; they are here held as the material's prisoners within a narrow circle

of vision; they are, so to speak, the subject's serfs, who may not leave the definite workplace. This intellectual slavery of the concepts forms only a one-sided counterweight to the equally one-sided unboundedness that proceeds from superficial abstractions, and serves the ingenious play of the association of ideas. Every concept that is had completely is had with freedom; it then depends on seeing wherein the freedom consists.

That the concept of the mechanical belongs just as originally under mechanics as the concept of the chemical under chemistry is incontrovertible; when now Hegel in his *Logic* nonetheless deals with the concept's own mechanism and its logical chemism, then it is neither mechanics's nor chemistry's, but rather philosophy's task to investigate with what right such a transfer has happened. Word-explanations, acute interpretations of homonyms, synonyms, metaphors, and so on, do not mean much in this case. Here the question is not of word-explanations; here the question is expressly of the changes in the categories' validity that are brought about by a passage through different spheres. Such investigations demand expertise and Dialectic. In that a concept has passed over from one sphere to another, it is no longer the same, and yet still the same: "function" in mathematics and "function" in physiology is not the same, and yet the same; "approximation" in morals and "approximation" in mathematics is not the same, and yet the same; "reflection" in optics and "reflection" in logic is not the same, and yet the same. Here one can continue to infinity.

Philosophy's humanizing activity does not immediately coincide with a striving toward the popular. The dialectical liberation of the concept, which at every point must be exactly conditioned by what is given by assimilation, belongs exclusively to science. This humanizing activity, a necessary condition for consciousness's being able in all things to be itself, and to master its many rubrics, is possible only in that philosophy steps forth independently, and lifts its own method.

3) *Inwardizing: a totalizing activity.* What difference there is between knowing something when consciousness especially enters into the learned rubric, and knowing something in such a way that on a given occasion it instantly presents itself, and stands clear to thought, is generally known. The same content of knowledge can in the one form fall so heavy that each time a peculiar exertion of force is required when the knower is to lift it; in another form, when it, without losing its intellectual specific gravity, has by assimilation been separated from the all-too-circumstantial admixtures,

it can on the contrary be so light that consciousness, freed from the material's burden, bears its wealth as though it bore only itself: *omnia mea porto mecum* [all that is mine I carry with me].

The totalizing activity is conditioned by a corresponding method. To the finite separation of thinking and being corresponds a *critical* method; reflection moves about its object, points out the dogmatic proposition's untenability, and makes a subjective transition from the one to the other. Thought moves, but the thing stands still; here one thinks about the object, but the object does not come to unfold its moments: here the immanent Dialectic of the development of the matter is lacking. To Idealism's absolute knowledge corresponds a *dialectical* method. Here it is too little to think about the matter, for the matter is thought through in such a way that it looks as though it thought itself. The objective then does not properly express the real; it is an abstraction in which the subjective comes objectively to itself. The objective is not an existing object, it is a concept; it goes by "der Begriff" [the concept] and "der Begriff"; the pure concept entertains itself: here criticism is lacking. To a synthetic unity of thinking and being, a unity of essence that cannot suspend the real difference, corresponds a method in which objective development of the matter and subjective reflection determine each other mutually; such a method is *critical-dialectical*: criticism clarifies the finite conditions, Dialectic opens up the inner infinity.

The methodical movement is in philosophy a double one. On the one side the special in the content, the finite with its particulars, is worked up and assimilated in such a way that the moments go back into the universal, that the fundamental cognition is enriched, and great totalities are set in motion. In this respect the Hegelian method has an imposing superiority. But since the real, with its finite members, its angular forms, and existential punctuality, necessarily belongs to the development of the matter, it is on the other side an equally essential task to analyze the individual, separate the members, penetrate the entanglements, and bring the individual to dialectical vibration. In this respect the Kierkegaardian analyses furnish apt examples. On the heights of totality the telescope is applied, for immersion in the analysis of punctuality the microscope is used.

In the outer world two such opposed modes of viewing cannot possibly be applied simultaneously; to look at once through a telescope and a microscope is doubtless an impossibility. In knowledge's inner world the matter stands somewhat otherwise. If

already the subject-science, with its whole detail and its rubrics, can work itself up to higher reciprocal relations of insight into the individual and survey of the whole, then philosophy, with its assimilating power and liberating activity, must be able to carry it still further. The critical conditions the dialectical. The more dialectical insight and survey — insight into the special and survey of the total, insight into the total and survey of the special — alternate with each other, and in the dialectical refraction awaken each other, sharpen each other, and penetrate each other, the greater is the intellectual tension, the more the concept's force grows, the higher the waves of the movement beat, the stronger the stream of cognition goes.

Method relates to system as vital activity to the living organism. In the subject-science, where it always depends on getting the material placed, the schema can play a role. In philosophy, which is not burdened by material, schematism is intolerable. Every principled thought is at bottom always systematic; if the systematic form will not at once present itself, the thought ought to be developed with freedom; it will then afterward be seen from the development in what comprehensive connection it belongs.

The finite knowledge has many particular points of departure and many particular systems; the "absolute" knowledge will develop all from one principle in one continuous system. The synthetic knowledge, which can neither be satisfied by relative points of departure, nor reach the absolute principle, gets no absolute system, but within its bound an essential relation of relative systems.

The more the individual elements of knowledge are worked together, organized, and clarified, the more intimate the connection between the whole and its individual parts becomes, the more intimately also the content of knowledge goes together with the knowing consciousness. The assimilating and humanizing activity that is here demanded shows clearly that the complete inwardizing is dependent on an independent philosophy.

If we grasp the particular sciences as definite organs in the great organism of scientificity, then we must evidently grasp philosophy as the soul of the organism. Here too it will then repeat itself, that a rising separation and more definite development of the individual organs goes together with a corresponding concentration. The principled unity of the functions will here too become more independent, according as it becomes more intimate, and, though in the finite sense a product of the organs'

cooperation, nonetheless with inner infinity assert a subsistence in itself over against the particular organs.
